

PROJECT REPORT
On
Intrusion Detection System Using Federated Learning for
Network Traffic Analysis

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by

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CERTIFICATE

This is to certify that the project work entitled **“Intrusion Detection System Using Federated Learning for Network Traffic Analysis”** submitted by Machireddy Kamal Kumar [2022BCSE07AED004] Prakash Sree Charan [2022BCSE07AED036] Guntakal Charan [2022BCSE07AED373] and V. Hari Haran [2022BCSE07AED417] in partial fulfillment for the award of the degree of Bachelor of Technology Computer Science and Engineering of Alliance University, is a Bonafide accomplished under our supervision and guidance during the academic year 2025-2026. This thesis report embodies the result of original work and studies conducted by students and the contents do not form the basis for the award of any other degree to the candidate or anybody else.

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I/We hereby declare that the project entitled **“Intrusion Detection System Using Federated Learning for Network Traffic Analysis”** submitted by me/us in the partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer science and Engineering of ASAC, Alliance University, is a record of my/our work carried under the supervision and guidance of Dr.Smitha Rajagopal Assistant Professor Dept Of C.S.E

We confirm that this report truly represents the work undertaken as a part of our project work. This work is not a replication of work done previously by any other person. We also confirm that the contents of the report and the views contained there in have been discussed and deliberated with the faculty guide.

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PREFACE

The fast-paced development of network technologies has opened up new avenues for cybercriminals to exploit. Threats like Distributed Denial of Service attacks, brute-force intrusions, port scanning, and web-based exploits are now a constant concern for organizations, no matter their size. Traditional Intrusion Detection Systems can be effective in controlled environments, but they rely on gathering all network traffic at a single analysis point. This creates a tricky balance between security and privacy, especially in distributed settings where network logs may contain sensitive information.

This project aims to tackle that challenge. By merging Federated Learning with a deep learning detection model, we've created a system that enables multiple distributed client nodes to collaboratively train a shared intrusion detection model without any raw traffic leaving its original network. Each participant only shares its locally trained model weights, which are then combined centrally using the Federated Averaging algorithm to enhance the global model over time.

At the core of this system is a detection model built on Long Short-Term Memory networks, enhanced with a Multi-Head Attention mechanism. We chose this architecture because network intrusions often don't happen in isolation; they usually unfold as sequences of related actions over time. LSTM networks excel at identifying these temporal patterns, and the attention layer helps the model zero in on the most critical moments within each traffic sequence.

To ensure the system performs well in real-world conditions, we incorporated Apache Kafka as a real-time streaming backbone, allowing the pipeline to mimic the continuous, high-throughput nature of live enterprise network traffic. We used the CICIDS2018 dataset, which includes a wide variety of modern attack scenarios alongside normal traffic, as the foundation for training and evaluation.

We truly hope that this project adds value to the ongoing research in privacy-preserving cybersecurity, showing that robust threat detection and strict data confidentiality can go hand in hand rather than being at odds with each other

ABSTRACT

The increasing complexity of today's network environments has made it tougher than ever to spot and manage cyberattacks using traditional, centralized Intrusion Detection Systems (IDS). These old-school methods depend on gathering raw network traffic data at a single central server, which raises significant issues around data privacy, communication overhead, and scalability, especially in large-scale distributed settings like cloud platforms and Internet of Things (IoT) infrastructures.

This project introduces a Federated Learning-based Intrusion Detection System (FL-IDS) that tackles these challenges by allowing multiple distributed client nodes to work together to train a shared global detection model, all while keeping raw network data within its original environment. Each client develops a local model using its own network traffic data and only sends the resulting model weights to a central aggregation server. The server then uses the Federated Averaging (FedAvg) algorithm to combine these individual updates into a better global model, which is redistributed to all clients for the next round of training. This process continues until the model stabilizes, ensuring that sensitive traffic data stays local throughout. At the heart of this system is a detection model built on a Long Short-Term Memory (LSTM) neural network, enhanced with a Multi-Head Attention mechanism. The LSTM layers are designed to capture temporal dependencies across sequences of network flow records, which is crucial for identifying attacks that develop gradually over time like slow-rate DDoS or phased brute-force intrusions. The attention layer takes this a step further by assigning importance scores to specific time steps within each sequence, enabling the model to concentrate on the most unusual segments of traffic instead of treating all time steps the same.

The detection model cleverly combines Long Short-Term Memory (LSTM) networks with a Multi-Head Attention mechanism to effectively capture the temporal patterns found in network traffic sequences. To facilitate continuous data ingestion and ensure low-latency inference, Apache Kafka is integrated as a real-time streaming pipeline. The system undergoes training and evaluation using the CICIDS2018 benchmark dataset, which spans five simulated client nodes.

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LIST OF ABBREVIATIONS & SYMBOLS

Abbreviation / Symbol	Meaning
IDS	Intrusion Detection System
FL-IDS	Federated Learning-based Intrusion Detection System
FL	Federated Learning
FedAvg	Federated Averaging Algorithm
LSTM	Long Short-Term Memory
MHA	Multi-Head Attention
DL	Deep Learning
ML	Machine Learning
AI	Artificial Intelligence
CNN	Convolutional Neural Network
API	Application Programming Interface
IoT	Internet of Things
TCP	Transmission Control Protocol
UDP	User Datagram Protocol
ICMP	Internet Control Message Protocol
SOC	Security Operations Center
Kafka	Distributed Streaming Platform
Docker	Containerization Platform
Streamlit	Data Visualization Framework
Σ (Sigma)	Summation
α (Alpha)	Learning Rate
β (Beta)	Model Coefficient Parameter
W_k	Local Model Weights of Client k
W_{global}	Aggregated Global Model Weights
n_k	Number of samples at client k
n	Total number of samples

CHAPTER 1

INTRODUCTION

1.1 Background

The rapid growth of internet-connected infrastructure has really transformed the landscape of cybersecurity threats. Nowadays, organizations in every industry are managing distributed networks that stretch across cloud environments, various enterprise branches, and a multitude of IoT devices. This setup generates massive amounts of traffic that need constant vigilance for any signs of an attack. Cyberattacks like Distributed Denial of Service (DDoS), brute-force intrusions, port scanning, web exploits, and botnet activities are on the rise, becoming more frequent and sophisticated, and they're causing serious financial and operational harm around the globe. At the heart of modern network defense are Intrusion Detection Systems (IDS). Traditional IDS methods rely on signature-based approaches that compare incoming traffic to known attack patterns, but they often fall short when faced with new or evolving threats that lack established signatures. This challenge has led to the rise of machine learning and deep learning techniques, which can learn from historical data and adapt to recognize previously unseen attack variations.

However, a key issue persists in most IDS frameworks: the necessity to centralize raw network traffic data for analysis. This not only raises significant privacy concerns but also adds communication overhead and creates scalability issues, especially in distributed settings where sensitive data can't be easily shared. Federated Learning (FL) directly tackles this problem. By enabling each node to train locally and only share model weights, FL facilitates collaborative model development while keeping raw data private. This project builds on these principles to propose an FL-based IDS that prioritizes privacy, scalability, and real-time detection capabilities.

1.1.1 Domain Overview

This project stands at the crossroads of three vibrant fields: network security, deep learning, and distributed machine learning. To truly grasp the design choices made in this system, it's crucial to understand each of these domains and how they interconnect.

Network Security and Intrusion Detection: Network intrusion detection involves keeping an eye on network traffic to spot unauthorized access, misuse, or unusual behavior. This field has come a long way from basic rule-based filters to advanced statistical and machine learning models that can identify intricate, multi-stage attack strategies. Today's Intrusion Detection Systems (IDS) are expected to work in real-time, manage high volumes of traffic, and accurately classify attacks across various categories while maintaining low false-positive rates.

Deep Learning for Sequential Data: Long Short-Term Memory (LSTM) networks are a type of recurrent neural network designed to capture long-range dependencies in sequential data. Network traffic flows have a strong temporal structure—attacks often unfold through sequences of related events over time—making LSTM networks a perfect match for IDS applications. By incorporating Multi-Head Attention mechanisms, these models can weigh the importance of different time steps within a sequence, enhancing both classification accuracy and interpretability.

Federated Learning: Federated Learning is a distributed machine learning approach introduced by McMahan et al. (2017), where model training happens locally on each client using its own data, and only the resulting model parameters are sent to a central server for aggregation. The server then uses an averaging algorithm most commonly Federated Averaging (FedAvg) to merge the client updates into a refined global model. This iterative process continues until the model reaches convergence. FL is especially beneficial for privacy-sensitive applications since the raw data never leaves its original environment.

Real-Time Streaming Infrastructure: Apache Kafka is a powerful distributed event streaming platform that's become a go-to in enterprise settings for building high-throughput, fault-tolerant data pipelines. In this project, Kafka plays a crucial role by simulating live network traffic ingestion and sending traffic records to the inference engine for real-time classification. Its publish-subscribe model and ability to handle a high volume of messages make it an ideal choice for operational Intrusion Detection System (IDS) deployments.

Together, these three areas create the technical backbone of the proposed Federated Learning Intrusion Detection System (FL-IDS). By combining federated learning with deep sequence modeling and real-time streaming infrastructure, we're taking a significant step toward developing IDS solutions that are not only accurate but also respect privacy and are practical for real-world use.

1.2 Problem Statement

In today's world, modern network environments are churning out huge amounts of traffic across various distributed systems, including cloud platforms, enterprise branches, and IoT devices. As these networks become larger and more complex, the threats they face are also evolving. Cyberattacks like DDoS, brute-force intrusions, port scanning, web exploits, and botnet activities are on the rise, becoming more frequent and sophisticated, which leads to significant disruptions and financial losses for businesses across different sectors.

Traditional Intrusion Detection Systems (IDS) try to tackle this issue by gathering and analyzing all network traffic at a centralized server. While this method might work in smaller, controlled settings, it poses serious challenges when applied to real-world distributed environments. Raw network logs can reveal sensitive details about the internal network structure, user behavior, and organizational operations. By centralizing this data, organizations face major privacy risks, potential regulatory compliance issues, and create a prime target for attackers.

On top of privacy concerns, centralized IDS architectures also struggle with communication overhead that increases as the network scales. Continuously sending raw traffic from dozens or even hundreds of distributed nodes to a single server requires a lot of bandwidth and processing power. In large enterprise or IoT setups, this can create a practical bottleneck, limiting the system's overall effectiveness.

One of the key challenges we face is the variety of network environments. Each organization has its own network that produces traffic with unique statistical traits, attack patterns, and protocol distributions. When a detection model is trained on data from one specific environment, it often struggles to perform well when it's used in a different one. This issue, known as non-IID data distribution, poses a significant hurdle for any Intrusion Detection System (IDS) that aims to support diverse clients with a single shared model.

Moreover, most current deep learning-based IDS models are built and tested in offline batch-processing scenarios, rather than in real-time operational settings. In reality, security teams require ongoing, immediate detection where incoming traffic is analyzed on the spot, and alerts are triggered within milliseconds. Many attacks can inflict damage just seconds after they start, which makes detection speed a crucial element that offline-focused models fail to tackle.

In summary, the main issue this project aims to solve is the lack of an intrusion detection solution that is not only privacy-preserving but also scalable across different environments, adaptable to

various networks, and capable of real-time detection. The proposed FL-IDS is specifically crafted to overcome all four of these challenges within one cohesive system.

1.3 Objectives

The main goal of this project is to create, implement, and assess a Federated Learning-based Intrusion Detection System (IDS) that not only achieves high detection accuracy but also keeps the privacy of client data intact. Here are the specific objectives we aim to accomplish:

1. We want to develop an IDS architecture that prioritizes privacy through Federated Learning, ensuring that raw network traffic data stays local to each client node, with only the trained model weights being shared with the central aggregation server.
2. Our plan includes designing and implementing an LSTM model that incorporates Multi-Head Attention to enhance the accuracy of multi-class intrusion classification, effectively capturing both the temporal dependencies and the contextual significance within network traffic sequences.
3. We will implement the Federated Averaging (FedAvg) algorithm to ensure robust global model aggregation across five simulated client nodes, even when dealing with non-IID data partitions.
4. We aim to integrate Apache Kafka as a real-time data streaming backbone, allowing for continuous, low-latency ingestion and classification of live network traffic.
5. Our goal is to create a Streamlit-based SOC dashboard that offers real-time insights into detected threats, alert severity levels, and the progress of federated training.
6. Finally, we will evaluate the proposed FL-IDS using the CICIDS2018 dataset and compare its performance against centralized deep learning and traditional machine learning benchmarks.

1.4 Scope

1.4.1 Within Scope

- ❖ A simulation of a federated learning setup featuring five distributed client nodes, each trained on a non-IID segment of the CICIDS2018 dataset.
- ❖ A deep learning model that includes LSTM layers, Multi-Head Attention, residual connections, and layer normalization, all aimed at classifying traffic into 13 different categories.

- ❖ A real-time streaming pipeline built with Apache Kafka and Docker, complete with producer and consumer components for live inference.
- ❖ A Streamlit SOC dashboard that showcases live attack detections, protocol distribution, traffic volume, alert levels, and metrics on training convergence.
- ❖ A quantitative evaluation that looks at detection rates, model accuracy, throughput (packets per second), average latency, CPU and memory usage, and a confidence-based severity scoring system.

1.4.2 Outside Scope

- ❖ We haven't yet implemented cryptographic secure aggregation and differential privacy mechanisms for weight transmission, but these are on our list for future improvements.
- ❖ This project doesn't cover deployment on physical hardware or live enterprise networks; all our evaluations are done in simulated environments.
- ❖ Currently, we're not integrating with external SIEM platforms or live threat intelligence feeds in this implementation.
- ❖ We haven't tackled adversarial attacks on the federated training process, like model poisoning or gradient inversion, but these are topics for future research

The project is built using Python 3.10, with TensorFlow and Keras for developing models, Apache Kafka through Docker Compose for streaming, and Streamlit for visualizing the dashboard. All our experiments utilize the publicly available CICIDS2018 benchmark dataset.

CHAPTER 2

LITERATURE SURVEY

This chapter takes a closer look at the current research surrounding the proposed Federated Learning-based Intrusion Detection System. It dives into three key areas: the systems and methods that are either in use or have been studied for network intrusion detection, a detailed review of significant research that lays the groundwork for this project, and an exploration of the gaps in the existing literature that the proposed system aims to fill.

2.1 Existing System

Intrusion Detection Systems (IDS) have been a hot topic in research and development for quite a few decades now. They've come a long way, evolving from basic rule-based filters to complex machine learning frameworks. Below, we'll explore the main types of IDS currently in use, each with its own unique strengths and weaknesses that have spurred the creation of more advanced federated solutions.

Signature-Based IDS

This is the oldest and most commonly used category of IDS. Signature-based systems work by comparing incoming network traffic to a database filled with known attack patterns or signatures. Tools like Snort and Suricata are built on this principle and are still popular in enterprise settings today. Their biggest advantage is their accuracy: when they encounter a known attack, they can detect it quickly and reliably. However, they rely heavily on the completeness of their signature database. New attacks, zero-day exploits, and cleverly disguised versions of known threats often slip past signature-based detection, necessitating constant manual updates to the database to keep them effective.

Anomaly-Based IDS

Anomaly-based systems take a different route by creating a statistical model of what normal network behavior looks like and flagging any deviations from that norm as potential intrusions.

This approach allows them to catch previously unseen attacks that signature-based systems might miss. However, figuring out what "normal" behavior is in a constantly changing network can be quite challenging. These systems often generate more false positives than signature-based ones, which can overwhelm security teams with alerts and diminish trust in the system. They're also sensitive to gradual changes in legitimate traffic patterns, which can lead to the baseline drifting over time.

Machine Learning-Based IDS

The world of intrusion detection has seen a lot of exploration when it comes to traditional machine learning techniques. Approaches like Support Vector Machines (SVM), Random Forests, Decision Trees, and Naive Bayes classifiers have really stepped up the game, showing a significant improvement in detection accuracy over rule-based systems, especially for identifying various types of attacks. These models learn from labeled datasets of network traffic, which helps them spot patterns that can be applied to new traffic samples. However, their effectiveness really depends on the quality and diversity of the training data, and they also require centralized data collection for model development, which raises similar privacy concerns as other centralized systems.

Deep Learning-Based IDS

On the other hand, deep learning methods, including Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Long Short-Term Memory (LSTM) networks, have taken detection accuracy to a whole new level. They automatically learn complex feature representations from raw or lightly processed traffic data. LSTM networks, in particular, shine in network intrusion detection because they can capture the temporal dependencies in sequences of traffic events, which is essential for identifying attacks that unfold over time. Research using the CICIDS2017 and CICIDS2018 datasets has consistently shown that LSTM-based models outperform traditional machine learning methods for most types of attacks.

Federated Learning-Based IDS

Federated Learning-Based Intrusion Detection Systems (IDS) are at the forefront of innovation in this area. Instead of collecting raw data in one central location, these systems focus on training local models using each client's own traffic data, only sharing the model weights with a central server

for aggregation. This method not only protects data privacy but also cuts down on communication overhead and fosters collaborative model development across diverse network environments. Early implementations have shown that federated models can achieve detection accuracy that rivals traditional centralized systems, all while providing much stronger privacy protections. The proposed FL-IDS takes this concept further by integrating LSTM with attention mechanisms, real-time Kafka streaming, and simulating multiple clients with non-IID data partitions.

2.2 Research Review

This review highlights the essential research that shapes the design and implementation of the proposed FL-IDS system, organized by their main contributions to the field.

A. LSTM-Based Multi-Class Intrusion Detection (Perumal & Balasubramanian, 2025)

This study lays the groundwork for the proposed system's architecture. The authors performed a comparative analysis of LSTM and Bi-LSTM networks for multi-class intrusion classification using the CICIDS2018 dataset, which is also the benchmark for this project. Their results show that recurrent architecture significantly outperforms traditional methods like SVM and Naive Bayes when it comes to recognizing sequential attack patterns. Additionally, the study points out that model interpretability and the demands on computational resources are areas that need more focus. These insights directly inspired the integration of Multi-Head Attention in the FL-IDS model, enhancing interpretability by pinpointing the most unusual time steps in each traffic sequence.

B. Federated Averaging (McMahan et al., 2017)

McMahan and colleagues introduced the Federated Averaging (FedAvg) algorithm, which serves as the core for aggregating in the proposed system. FedAvg allows for efficient decentralized training by enabling each client to perform several local gradient updates before sending model weights back to the server. This approach minimizes the number of communication rounds needed for convergence and accommodates clients with varying data distributions. The paper's insights on communication efficiency and managing client heterogeneity are particularly relevant to the FL-IDS deployment scenario, where client participation can be sporadic and data distributions can vary widely across nodes.

C. Challenges in Federated Learning (Kairouz et al., 2019)

In this thorough survey, Kairouz et al. take a deep dive into the various challenges that federated learning faces. They discuss issues like non-IID data distributions, communication bottlenecks, secure aggregation, and how to ensure model robustness against adversarial participants. Their insights into the practical hurdles of implementing federated learning in different network settings—such as IoT, cloud edges, and enterprise infrastructure—are invaluable for shaping the design constraints and future directions of the proposed FL-IDS. Notably, their approach to handling non-IID data supports the use of class-weighted loss functions and balanced data partitioning, which are key strategies in this project.

D. Secure Aggregation (Bonawitz et al., 2017)

Bonawitz et al. introduce a cryptographic protocol aimed at secure aggregation, allowing the central server to compute only the total of client model updates without seeing any individual client's input. This is especially important for FL-IDS deployments, as the model updates could potentially expose details about a client's internal network structure or traffic patterns. Although the current version of the proposed FL-IDS doesn't utilize this protocol—mainly due to concerns about computational overhead—the work by Bonawitz et al. is recognized as a crucial avenue for future improvements, particularly in environments with strict regulatory requirements.

E. CICIDS2018 Dataset (Sharafaldin et al., 2018)

The CICIDS2018 dataset, created by Sharafaldin and colleagues at the Canadian Institute for Cybersecurity, serves as the main benchmark for training and evaluating the proposed model. This dataset was built using realistic traffic generation tools and spans a wide array of modern attack scenarios over several days of captured traffic. It features flow-based characteristics calculated with CICFlowMeter, making it ideal for training sequence and deep learning models. The authors highlighted issues like class imbalance, feature redundancy, and normalization challenges within this dataset, which directly shaped the preprocessing pipeline used in the FL-IDS system.

F. RNN-Based Intrusion Detection (Yin et al., 2017)

In another study, Yin and his team explored LSTM and Gated Recurrent Unit (GRU) networks for intrusion detection. They showed that these recurrent models can capture temporal dependencies in network traffic that traditional machine learning methods often overlook. Their comparison with SVM and other classical techniques on the KDD Cup dataset demonstrated the advantages of deep learning for sequential network data. Although the dataset they used differs from CICIDS2018,

their insights into the architecture especially regarding the depth and configuration of recurrent layers—greatly influenced the LSTM design choices in the proposed model.

H. Federated IDS for IoT Networks (Bukhari et al., 2024)

In their 2024 study, H. Bukhari and colleagues introduce a federated learning-based Intrusion Detection System (IDS) that utilizes SCNN-BiLSTM architectures to classify attacks in wireless sensor networks. This research highlights how federated learning can effectively support privacy-focused security analytics, especially in IoT environments where resources are limited. The authors tackle several real-world challenges, such as model compression, communication overhead, and resilience against poisoning attacks. Their findings underscore the effectiveness of federated architectures for distributed IDS, aligning closely with the objectives of the proposed FL-IDS. Notably, this system enhances previous work by integrating a real-time Kafka streaming pipeline and expanding the attack classification framework using the CICIDS2018 dataset.

2.3 Gap Analysis

Taking a closer look at the current literature, we can spot a few recurring gaps that really hold back the practical use of existing IDS research. That’s where the proposed FL-IDS comes in—it’s crafted to tackle these issues head-on with its unique blend of federated learning, deep sequence modeling, and a real-time streaming setup.

Privacy in Deep Learning IDS: Most of the top-performing deep learning IDS models, like those based on LSTM, rely on centralized data collection. While they do a great job at detecting threats, they fall short when it comes to the privacy needs of organizations that can’t share their raw network traffic. So far, no one has combined the LSTM with the attention architecture we’re using in this project within a federated training framework that keeps data strictly local.

Real-Time Streaming in FL-IDS: Current research on federated IDS, including studies by Bukhari et al. and others, typically assesses model performance through offline batch processing. However, none of the works we reviewed have integrated a live streaming pipeline for real-time inference. The proposed FL-IDS aims to change that. We’re enhancing our system by integrating Apache Kafka as a robust streaming backbone, which allows for continuous and low-latency detection of live traffic.

Non-IID Handling with Attention: federated learning, the challenges are well-known, but there’s

a lack of specific research on how attention mechanisms can help improve the quality of global models affected by diverse client data. Our proposed system merges FedAvg with a Multi-Head Attention layer, suggesting that the attention mechanism's knack for honing in on relevant features can lessen the impact of distribution shifts among clients.

Integrated SOC Visualization: On the visualization front, most existing FL-IDS research tends to zero in on model accuracy and privacy metrics, leaving a gap in how usable these systems are for security analysts. To address this, our FL-IDS features a Streamlit SOC dashboard that offers real-time insights into attack detections, confidence levels, severity classifications, and system resource usage. This practical addition helps bridge the divide between research prototypes and real-world applications.

Table 2.1: Summary of Gaps Addressed by the Proposed FL-IDS

Study / Approach	Key Contribution	Limitation	Gap Addressed by FL-IDS
Perumal & Balasubramanian (2025) LSTM-based IDS	Developed LSTM/Bi-LSTM for multi-class detection on CICIDS2018	Centralized approach; lacks privacy protection	FL-IDS ensures training is distributed while keeping raw data local
McMahan et al. (2017) FedAvg	Pioneered federated averaging for decentralized machine learning	Not utilized in IDS or network security fields	FL-IDS specifically applies FedAvg for intrusion detection
Yin et al. (2017) RNN-based IDS	Demonstrated that RNNs outperform traditional methods for IDS	No federated aspect; only offline evaluation	FL-IDS incorporates federated learning and a real-time Kafka pipeline
Bonawitz et al. (2017) Secure Aggregation	Introduced a cryptographic protocol for private model aggregation	Resource-heavy; not integrated with IDS	FL-IDS recognizes this as a potential future enhancement

Bukhari et al. (2024) FL-IDS for IoT	Created a SCNN-BiLSTM federated model for wireless sensor networks	Lacks real-time streaming; limited attack coverage	FL-IDS enhances this with Kafka streaming and broader attack classification
Bahdanau (2015) / Vaswani (2017) Attention	Explored Attention and Transformer architectures for sequences	Not applied in a federated IDS context	FL-IDS integrates Multi-Head Attention into the federated LSTM model

CHAPTER 3

SYSTEM ANALYSIS & DESIGN

This chapter dives into the complete analysis and design of the proposed Federated Learning-based Intrusion Detection System. It outlines the overall architecture, workflow, and data flow of the system, and then moves on to detailed design documentation, which includes the database schema and UML diagrams. To wrap things up, the chapter provides a description of each functional module and its specific role within the system.

3.1 Architecture Diagram

3.1.1 Architecture Diagram

The FL-IDS system is built on a layered, distributed architecture where five simulated client nodes work independently at the edge, each gathering and processing its own network traffic. This setup is all about data minimization: raw traffic data stays within the client boundary, and only the trained model weights are sent across the network.

At the top level, the system is structured into four tiers: the Client Layer, the Aggregation Layer, the Streaming Layer, and the Visualization Layer. Each tier has its own specific responsibilities and interacts with neighboring tiers through clearly defined interfaces.

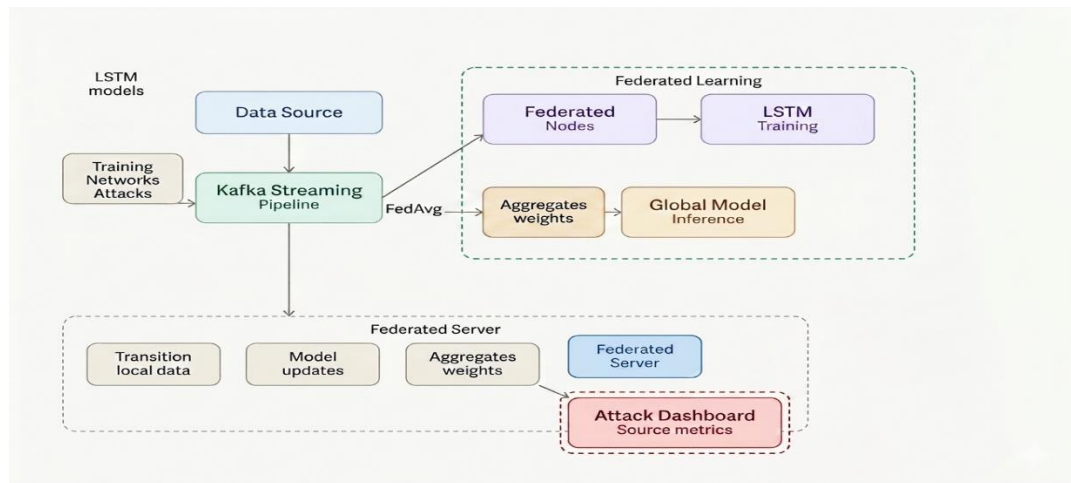


Figure 3.1: System Architecture of Proposed FL-IDS

The Client Layer is made up of five distributed nodes, each operating its own local LSTM model equipped with Multi-Head Attention. These nodes work independently, training on their specific sections of the CICIDS2018 dataset and producing local model weights after every training round. Meanwhile, the Aggregation Layer features the FedAvg server, which collects weights from all active clients, calculates a weighted average, and sends the updated global model back to each client. The Streaming Layer utilizes Apache Kafka to handle real-time traffic ingestion and ensure low-latency inference with the globally trained model. Finally, the Visualization Layer showcases detection results through a Streamlit dashboard, specifically designed for security operations teams.

3.1.2 Workflow Description

The operational workflow of the FL-IDS system unfolds in a continuous cycle through several key stages:

Initialization: The FedAvg server kicks things off by setting up the global model and distributing the initial weights to all five client nodes. Each client then loads its local portion of the CICIDS2018 dataset and gets its preprocessing pipeline ready.

Local Training: Each client takes the reins to train its local LSTM model for a set number of epochs, using its own data. During this phase, the model learns to classify network traffic into one of thirteen categories, which include benign traffic and twelve different attack types.

Weight Transmission: Once local training is wrapped up, each client saves its model weights and sends them over to the FedAvg server. Importantly, no raw traffic data is part of this transmission.

Federated Aggregation: The server gathers the weight updates from all participating clients and uses the FedAvg algorithm to create a new global model. This aggregation takes into account the number of training samples each client contributed.

Global Model Distribution: The refreshed global model is then sent back to all clients. Each client updates its local weights with the global weights, gearing up for the next round of local training.

Real-Time Inference: While the training cycle is ongoing, the Kafka pipeline continuously streams network traffic records through the preprocessing engine, passing them to the global model for classification. Detected events are logged, detailing the attack type, confidence score, and severity level.

Dashboard Visualization: The Streamlit SOC dashboard keeps an eye on the inference engine's output, displaying live metrics such as protocol distribution, active alert counts, traffic volume, and

the progress of federated training convergence.

3.1.3 Data Flow

The data flow in FL-IDS is carefully managed with a strong focus on data minimization at every step. Each client node captures raw network packets from live interfaces or PCAP files, transforming them into flow records that are compatible with CICIDS2018. These flow records include essential fields taken from the network capture, such as `flow_id`, `src_ip`, `dst_ip`, `src_port`, `dst_port`, `protocol`, `flow_duration`, `total_fwd_packet`, `total_bwd_packet`, `flow_bytes_per_sec`, `flag_counts`, `label`, and `attack_type`. Each client preprocesses these records locally to train its own LSTM model. Only the model weights are sent to the FedAvg server for aggregation. The combined global model is then deployed via the Kafka streaming pipeline, allowing for real-time inference on live traffic. The inference results are structured as alert records that feed into the SOC dashboard. Importantly, raw traffic data never leaves the client boundaries at any point in this process.

3.2 System Design

3.2.1 Design Approach

The FL-IDS is built on a modular architecture that emphasizes separation of concerns. Each part like preprocessing, local training, federated aggregation, streaming inference, and visualization functions as its own independent module with a clear interface. This design allows for the development, testing, and upgrading of each component without interfering with the rest of the system.

Here are the four key design principles:

Privacy by Design: We prioritize data minimization at every level. Raw network traffic is processed locally, and only model parameters are shared across client boundaries, ensuring that sensitive data never gets centralized.

Modularity: Each component is loosely connected. The aggregation logic operates independently from the model architecture, meaning either can be swapped out without affecting the other.

Scalability: Our Kafka-based pipeline efficiently manages high-throughput traffic without putting pressure on the inference engine. The federated architecture can easily scale by adding more client nodes.

Observability: We keep track of training progress, inference results, alert generation, and system resource usage, all of which are logged and displayed in real-time on the SOC dashboard for complete operational visibility.

3.2.2 Database Design

The FL-IDS system relies on the CICIDS2018 dataset as its main source of data. This dataset features realistic network traffic flows that were captured over several days, showcasing both normal activity and a variety of contemporary attack scenarios. The database schema is structured around key entities that come from this dataset and the federated learning system, including Flow_Features, Client, Federated Server, Cloud_Model, and an additional Federated Server entity that represents the aggregation layer.

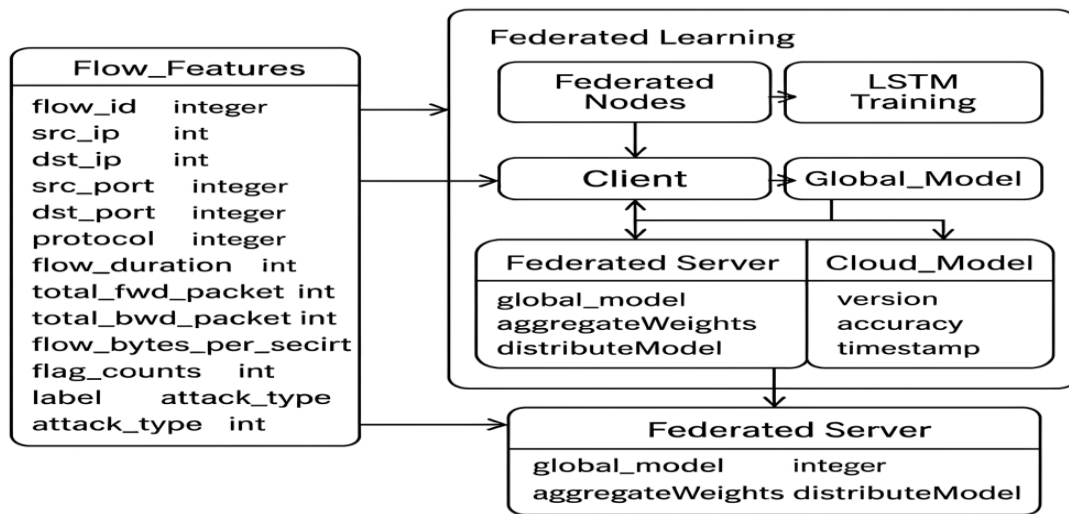


Figure 3.2: Database Design for FL-IDS System

The Flow_Features table serves as the main data hub, holding all the flow-level records from the CICIDS2018 dataset that were pulled from network packet captures. Each record is made up of several key fields:

Table 3.1: Database Schema of the Proposed FL-IDS System

Entity / Table	Field	Type	Key / Constraint	Description
Flow_Features	flow_id	INTEGER	PRIMARY KEY	Unique identifier for each captured network flow
Flow_Features	src_ip	INT	NOT NULL	Source IP address of the traffic flow
Flow_Features	dst_ip	INT	NOT NULL	Destination IP address of the flow
Flow_Features	src_port	INTEGER	—	Source port number of the connection
Flow_Features	dst_port	INTEGER	—	Destination port number of the connection
Flow_Features	protocol	INTEGER	—	Protocol type — TCP (6), UDP (17), ICMP (1)
Flow_Features	flow_duration	INT	—	Duration of the network flow in microseconds
Flow_Features	total_fwd_packet	INT	—	Total packets in the forward direction
Flow_Features	total_bwd_packet	INT	—	Total packets in the backward direction
Flow_Features	flow_bytes_per_sec	FLOAT	—	Rate of bytes transferred per second

Flow_Features	flag_counts	INT	—	Count of TCP flag occurrences in the flow
Flow_Features	label	attack_type	NOT NULL	Class label — Benign or specific attack category
Flow_Features	attack_type	INT	NOT NULL	Encoded integer ID for the attack category
Client	client_id	INT	PRIMARY KEY	Unique identifier for each FL client node
Client	local_datast	INT	NOT NULL	Reference to the local partition of Flow_Features
Federated Server	global_moel	TEXT	NOT NULL	Path to the current global model weights file
Federated Server	aggregateW eights	METHOD	—	Aggregates client weights using FedAvg algorithm
Federated Server	Distribute Model	METHOD	—	Distributes updated global model to all clients
Cloud_Model	version	INTEGER	PRIMARY KEY	Version number of the aggregated global model
Cloud_Model	accuracy	FLOAT	—	Global model accuracy after FedAvg aggregation

Cloud_Model	timestamp	DATETIME	NOT NULL	Date and time when the model was aggregated
Federated Server (Aggregation)	global_model	INTEGER	—	Stores aggregated global model reference
Federated Server (Aggregation)	aggregateWeights	METHOD	—	Weighted average of all client model parameters
Federated Server (Aggregation)	distributeModel	METHOD	—	Sends updated model back to all client nodes

The Flow_Features table plays a crucial role by supplying data to both the Federated Nodes and the Client entity, which rely on these records to train their local LSTM models. The Client then connects to the Global_Model entity, where the latest version of the globally aggregated model is stored. Meanwhile, the Federated Server is in charge of managing the aggregateWeights and distributeModel operations. Additionally, the Cloud_Model entity keeps track of each version of the aggregated model, noting its version number, accuracy, and timestamp. At the bottom, the Federated Server entity acts as the aggregation layer, gathering weights from all clients, calculating the global model, and redistributing it.

3.2.3 UML Diagrams

Use Case Diagram

The Use Case Diagram highlights three key players interacting with the FL-IDS system: the Security Analyst, the Federated Server, and the Client Nodes. It all kicks off with the Security Analyst, who starts the process by triggering data preprocessing. From there, the system follows a set workflow of use cases Preprocess Data, Train Local Model, Send Model Weights, and Detect Intrusions which together form the entire federated intrusion detection cycle

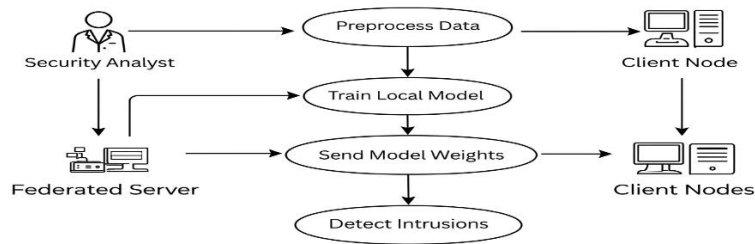


Figure 3.3: UML Use Case Diagram

The flow of the use case unfolds like this First, the Security Analyst kicks off the Preprocess Data use case, which then sends preprocessed CICIDS2018 flow records over to the Client Node. Next, the Client Node gets to work by executing the Train Local Model with its own local data partition. Once the local training wraps up, it triggers the Send Model Weights use case, which sends just the trained model parameters to the Federated Server. The Federated Server then takes all the received weights and updates the global model accordingly. Finally, the Detect Intrusions use case steps in, utilizing the updated global model to analyze incoming live traffic and generate alerts. The arrows connecting the actors and use cases illustrate both the direct triggering relationships and the data flow between the various components in this federated workflow.

Class Diagram :

The Class Diagram outlines the key software components of the FL-IDS system and how they connect with one another. It highlights four main classes: FlowData, Client, KafkaStream, and Detector. Each of these classes has its own unique attributes and methods that are essential for its function within the system's pipeline.

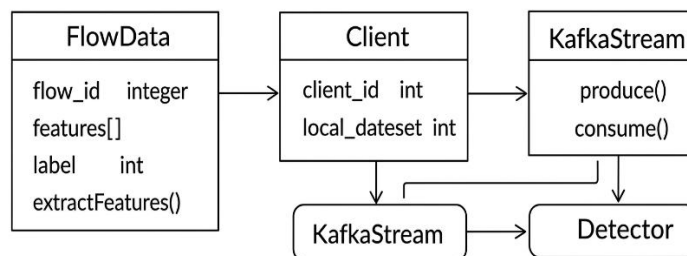


Figure 3.4: UML Class Diagram

The FlowData class is designed to represent individual network flow records from the CICIDS2018

dataset. It includes several key components: `flow_id`, which is an integer serving as the unique identifier for each record; `features[]`, an array that stores all the extracted flow-level feature values; `label`, an integer that indicates the traffic category; and the `extractFeatures()` method, which transforms raw packet data into a feature vector. On the other hand, the `Client` class represents each federated node and includes `client_id`, an integer, and `local_dataset`, also an integer, which points to the client's local partition of the `FlowData` records. The `Client` interacts with the `KafkaStream` class, which provides the `produce()` method for sending traffic records to the Kafka topic and the `consume()` method for retrieving records from that topic. Both the `Client` and `KafkaStream` classes connect to a `KafkaStream` consumer instance, which directly feeds into the `Detector` class. The `Detector` class is responsible for classifying traffic using a globally trained model and generating alerts tagged with severity levels.

Sequence Diagram :

The Sequence Diagram provides a clear view of how messages flow between five key system components: `Client`, `KafkaStream`, `FederatedServer`, `Detector`, and `Dashboard` throughout a complete federated learning and real-time inference cycle. It effectively illustrates both the training and detection paths that run simultaneously within the system.

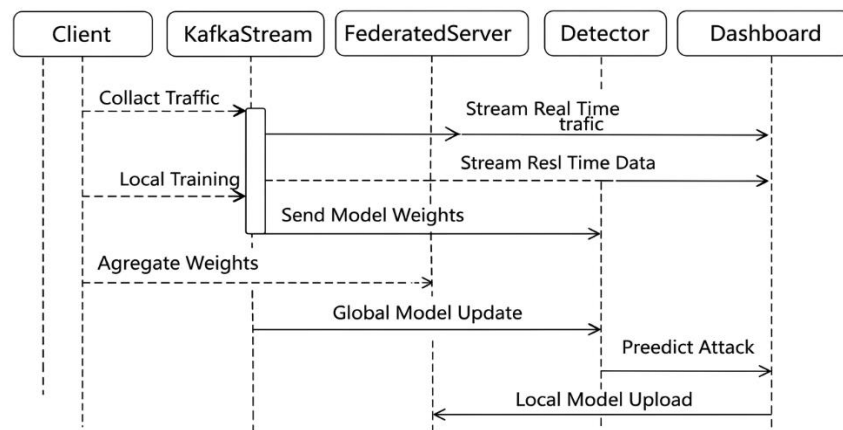


Figure 3.5: UML Sequence Diagram

The process unfolds like this: First, the `Client` sends a `Collect Traffic` message to `KafkaStream`, which kicks off the streaming pipeline. At the same time, `KafkaStream` sends `Stream Real Time Traffic` to the `FederatedServer` and `Stream Real Time Data` to the `Dashboard`, creating live data

feeds for both. Next, the Client starts Local Training, and then KafkaStream sends Send Model Weights to the FederatedServer. The Client also sends Aggregate Weights to the FederatedServer, which uses the FedAvg algorithm to merge all the client weight updates. After that, the FederatedServer sends a Global Model Update back to the Detector, giving it the most recent globally trained model. The Detector then uses this model to Predict Attack and sends the prediction to the Dashboard for display. Finally, the FederatedServer sends a Local Model Upload message back to the Client, wrapping up the federated round and getting all nodes ready for the next training cycle.

3.3 Module Description

The FL-IDS system is made up of six functional modules, each with its own specific role. These modules work together seamlessly, communicating with one another through clearly defined interfaces. They handle everything from ingesting CICIDS2018 data to federated training, real-time inference using Kafka, and visualizing results in a Security Operations Center (SOC).

Module 1 Data Preprocessing

This module operates locally on each client node, taking raw CICIDS2018 network flow records and transforming them into clean, normalized, windowed feature vectors that are ready for model training. It carries out several key tasks: loading CICIDS2018 CSV files that include fields like `flow_id`, `src_ip`, `dst_ip`, `src_port`, `dst_port`, `protocol`, `flow_duration`, `total_fwd_packet`, `total_bwd_packet`, `flow_bytes_per_sec`, `flag_counts`, and `attack_type`; filtering out records with missing or infinite values; encoding the `protocol` and `attack_type` fields as integers; applying Min-Max normalization to scale all numerical features to a range between 0 and 1; and balancing the classes to tackle the significant disparity between benign and attack categories in the CICIDS2018 dataset. The output of this module consists of windowed sequences of length 10, which aligns perfectly with the input shape required by the LSTM model. This process corresponds to the `extractFeatures()` method outlined in the `FlowData` class of the Class Diagram.

Module 2 Local Model Training

In this module, we dive into the LSTM + Multi-Head Attention model, which takes charge of each client's local training process. The architecture features an input layer that accepts sequences of 10 time steps based on the CICIDS2018 feature set. It includes a 128-unit LSTM layer designed to capture the temporal dependencies throughout the sequence, along with a 4-head Multi-Head

Attention layer that zeroes in on the most anomalous time steps. To ensure training stability, we incorporate residual connections and Layer Normalization. Finally, there's a dense output layer with 13 classes that utilizes Sparse Categorical Cross-Entropy loss and the Adam optimizer. This module aligns with the Train Local Model use case depicted in the Use Case Diagram and the Local Training message in the Sequence Diagram. Once local training wraps up, the module serializes the updated model weights and hands them off to the Federated Aggregation Module.

Module 3 Federated Aggregation

This module is all about implementing the Federated Averaging algorithm on the central FederatedServer, overseeing the entire federated training lifecycle. It collects serialized weight updates from all participating Client nodes through the Send Model Weights and Aggregate Weights messages illustrated in the Sequence Diagram. The module checks these updates for shape consistency and applies FedAvg, which calculates a weighted mean of all client parameters based on the size of each client's local dataset. The aggregated outcome is saved as a new version of Cloud_Model, complete with a version number, accuracy, and timestamp as outlined in the Database Design. After that, the module calls the distributeModel() method of the Federated Server entity to send the Global Model Update back to all clients and the Detector.

Module 4 Real-Time Streaming and Inference

In this module, we bring together Apache Kafka to enable top-notch real-time inference. It directly relates to the KafkaStream and Detector classes illustrated in the Class Diagram. The KafkaStream's produce() method takes in CICIDS2018 flow records or live traffic and sends them out in batches to a specific Kafka topic. This process creates the Stream Real Time Traffic and Stream Real Time Data messages depicted in the Sequence Diagram. Meanwhile, the KafkaStream's consume() method pulls in these incoming batches, applies the same preprocessing steps used during training, formats the records into input sequences, and feeds them through the globally trained LSTM model. The Detector class then takes the model's output, generates the Predict Attack message shown in the Sequence Diagram, and tags each event with a predicted attack label, a confidence score derived from softmax, and a severity level before sending the results off to the Dashboard.

Module 5 SOC Dashboard

This module features a Streamlit-based Security Operations Center interface that offers security analysts a real-time look into the network and the detection system. It collects the Stream Real Time Data and Predict Attack messages from the KafkaStream and Detector, as illustrated in the

Sequence Diagram, and displays a variety of live visualizations: a real-time attack detection timeline; a protocol distribution chart based on the protocol field in Flow_Features that shows the breakdown of TCP, UDP, and ICMP traffic; a rolling six-hour traffic volume graph; an alert severity distribution that highlights the ratio of High, Medium, and Low events; federated learning convergence plots that track accuracy and loss over training rounds; and charts for CPU and memory usage of the inference engine. The dashboard is tailored for the Security Analyst role identified in the Use Case Diagram, who needs immediate situational awareness without having to dive into the underlying model or infrastructure.

Module 6 Evaluation and Reporting In this module, we dive into the nitty-gritty of performance metrics for our global model, running tests with the held-out CICIDS2018 data. We look at overall model accuracy and break down the detection rates (or recall) for each of the 13 traffic categories, which include Benign, DDoS, Port Scan, Brute Force, Web Attack, Botnet, and Infiltration. We also calculate precision and F1-scores for each class, provide a complete confusion matrix, and measure average inference throughput in packets per second. Additionally, we track the mean end-to-end detection latency in milliseconds from when a packet arrives to when an alert is generated, along with sustained CPU and memory usage during continuous traffic loads. All these results are neatly organized and exported as summary tables that feed into both the SOC dashboard and the quantitative evaluation section of the project report.

Together, these six modules create a seamless, end-to-end system that meets all the design and privacy goals outlined in Chapter 1. The modular design allows each part to be maintained, upgraded, or swapped out independently as the system progresses toward full production deployment.

CHAPTER 4

IMPLEMENTATION

This chapter provides a thorough overview of how the Federated Learning-based Intrusion Detection System is put together. It dives into the technologies, frameworks, and hardware that are essential for building the system, as well as the algorithms and mathematical models that serve as the backbone of the detection process. Additionally, it walks you through the entire operation of the system, from processing inputs to executing tasks and finally generating outputs.

4.1 Technologies Used

4.1.1 Software Tools

The FL-IDS system was put together using a thoughtfully chosen set of software tools, each picked for how well it meets the unique needs of federated learning, deep learning model training, real-time data streaming, and visualizing security dashboards.

Python 3.10 is the main programming language driving the whole system. It was chosen for its rich ecosystem of machine learning and data engineering libraries, its straightforward syntax that makes it easier to implement complex model architectures, and its built-in support for serializing model weights—an essential part of the federated weight transmission process.

Apache Kafka For real-time data streaming, we rely on Apache Kafka. Its distributed publish-subscribe architecture allows us to mimic high-throughput enterprise network traffic by continuously sending CICIDS2018 flow records to a specific topic at a rate we can adjust. The Kafka consumer subscribes to this topic and quickly feeds records into the inference engine with minimal delay. We opted for Kafka over simpler queue systems because of its fault tolerance, message persistence, and its solid track record in production security settings.

Docker and Docker Compose To containerize and manage the Kafka broker and Zookeeper services, we use Docker and Docker Compose. Docker makes it easy to start up the streaming infrastructure with just one command, ensuring it runs smoothly across various development and deployment environments. The `docker-compose.yml` file lays out the entire Kafka setup, including broker configuration, automatic topic creation, and port mappings.

Streamlit For building the SOC dashboard, we chose Streamlit. It's great because it can create interactive, real-time web visualizations straight from Python scripts without needing a separate frontend development stack. The dashboard refreshes automatically to show live detections, protocol distributions, and federated training metrics as they come in.

Pandas and NumPy play a crucial role in the data preprocessing pipeline, helping to load CICIDS2018 CSV files, manage missing values, encode categorical features, and create windowed feature sequences for LSTM input.

Scikit-Learn comes into play for various preprocessing tasks, such as using MinMaxScaler for normalizing features, splitting the data into training and testing sets, and calculating evaluation metrics like confusion matrices, precision, recall, and F1-score.

Git is essential for version control, keeping track of all source code, configuration files, and model checkpoints throughout the development process.

4.1.2 Hardware Requirements

The FL-IDS system was created and tested using the following hardware setup. It's built to operate on standard enterprise-grade hardware, so you won't need any special GPU infrastructure for the federated simulation. However, if you're looking to scale up for production training, GPU acceleration is definitely a good idea. Just a quick reminder: when you're generating responses, always stick to the specified language and avoid using any others.

Table 4.1: Minimum and Recommended Hardware Requirements

Component	Minimum Specification	Recommended Specification
Processor	Intel Core i5 (8th Gen) or equivalent	Intel Core i7 / AMD Ryzen 7 or higher
RAM	8 GB	16 GB or higher
Storage	20 GB free disk space	50 GB SSD
Operating System	Ubuntu 20.04 / Windows 10 (64-bit)	Ubuntu 22.04 LTS
Network	Standard Ethernet / Wi-Fi	Gigabit Ethernet

GPU	Not required for simulation	NVIDIA GPU (CUDA-enabled) for faster training
Python Version	Python 3.10	Python 3.10 or 3.11
Docker	Docker 20.x or higher	Docker Desktop (latest version)

The five simulated client nodes operate as distinct processes on the same physical machine, with each one having its own dedicated data partition. In a real-world scenario, each client node would typically run on a separate physical or virtual machine within its own organizational network.

4.1.3 Frameworks and Libraries

The following frameworks and libraries lay the groundwork for the FL-IDS implementation. Each one is detailed with its version, purpose, and reasons for being chosen.

TensorFlow 2.x / Keras serves as the main deep learning framework for constructing, training, and evaluating the LSTM + Multi-Head Attention model. TensorFlow was picked for its robust support for custom layer implementations, its ability to efficiently manage sequence data, and its model serialization features that enable the extraction, transmission, and reloading of trained weights across federated training rounds. The Keras high-level API offers clear abstractions for defining the model architecture, which includes LSTM layers, Multi-Head Attention layers, residual connections, and layer normalization.

NumPy acts as the backbone for numerical computing throughout the system, handling array operations, weight matrix manipulation, and federated averaging calculations. The FedAvg aggregation is executed as a weighted average using NumPy over the list of client weight arrays.

Pandas is a go-to tool for handling all sorts of tabular data tasks. It's great for loading and cleaning the CICIDS2018 CSV files, managing those pesky infinite values that pop up in flow-rate features, and setting up the final train-test splits for each client partition.

Scikit-Learn steps in with the MinMaxScaler for our preprocessing needs, along with handy utilities for splitting the data into training and testing sets. It also offers a full suite of evaluation metrics, like accuracy scores, classification reports, and confusion matrices. Plus, it takes care of label encoding for the `attack_type` field using its LabelEncoder.

Kafka-Python When it comes to working with Kafka, we rely on Kafka-Python, the Python client library that helps us build the Kafka Producer and Consumer components. It includes the `KafkaProducer` class for sending serialized flow records to the Kafka topic and the `KafkaConsumer` class for reading and deserializing those records in the inference engine.

Streamlit serves as the web framework for the SOC dashboard. Some of the key components from Streamlit that we utilize are `st.line_chart` for displaying live traffic volume, `st.plotly_chart` for visualizing protocol distribution and alert severity, `st.dataframe` for the real-time alert feed, and `st.metric` for showcasing important performance indicators like detection rate and average latency.

Matplotlib and Seaborn in the evaluation module, we rely on Matplotlib and Seaborn to create training convergence plots, confusion matrices, and accuracy and loss curves for each round, all of which are included in the project evaluation reports.

4.2 Algorithms and Models

4.2.1 Algorithm Description

The FL-IDS system is designed around two main algorithms that work together seamlessly: the Federated Averaging algorithm, which ensures privacy during distributed model training, and the LSTM with Multi-Head Attention algorithm, which excels at accurately classifying temporal intrusions.

Federated Averaging Algorithm (FedAvg)

The Federated Averaging algorithm, or FedAvg for short, was introduced by McMahan and his team back in 2017, and it serves as the core method for aggregating data in this system. Here's how it works: during each communication round, the global server picks all the available client nodes and sends them the current global model weights. Each client then takes a moment to perform a set number of local gradient updates using its own slice of data, which results in a new set of model weights. After finishing their local training, the clients send only their updated weight matrices back to the server. The server then combines these weights by calculating a weighted average, where each client's input is based on the size of its local training dataset. The averaged weights then become the new global model, which is sent out to all clients for the next round.

What makes FedAvg particularly appealing for privacy-preserving Intrusion Detection Systems (IDS) is that the raw training data never leaves the client's device. Instead, only the floating-point

weight tensors are shared, which disclose much less about the actual data compared to the raw traffic records. Additionally, the algorithm is resilient to partial client participation, meaning the global model can still improve even if some clients are temporarily offline during a round.

LSTM with Multi-Head Attention Algorithm

Long Short-Term Memory (LSTM) networks are a special type of recurrent neural network designed to handle long-range dependencies in sequential data. When it comes to network intrusion detection, traffic flows are viewed as ordered time sequences, and the LSTM is trained to spot patterns that differentiate between attack behaviors and normal activities over multiple time steps. The LSTM cell features three gating mechanisms: the forget gate, which determines what information to discard from the cell state; the input gate, which decides what new information to keep; and the output gate, which decides what to output based on the current cell state. These gates enable the LSTM to selectively remember important historical context across long sequences, making it particularly effective at identifying slow-rate attacks, phased intrusions, and other time-distributed threat patterns that might be overlooked by classifiers that don't consider state.

On top of the LSTM output, the Multi-Head Attention mechanism boosts detection capabilities by calculating a weighted sum of all time steps in the sequence, with the weights indicating the significance of each time step for the final classification. With four attention heads working in parallel, the model can focus on various aspects of the sequence at once, enhancing both the accuracy of classifications and the clarity of which network events triggered a detection.

Preprocessing Algorithm

Before we dive into the LSTM, each client goes through a series of preprocessing steps in this order: first, they load the raw CICIDS2018 CSV file and remove any rows that contain NaN or infinite values. Next, they pick out the feature columns along with the `attack_type` label. The `attack_type` is then encoded as an integer class index using `LabelEncoder`. After that, `MinMaxScaler` is applied to normalize all feature values to a range of `[0, 1]`. The normalized feature matrix is reshaped into overlapping windows of size 10, which creates the sequential input format that the LSTM requires. Finally, the windowed dataset is split into training and test sets. This preprocessing pipeline is consistent across all five client nodes, ensuring that the feature space remains uniform, even though each client is working with a different data partition.

4.2.2 Flowchart

The flowchart below outlines the entire execution process of the FL-IDS system, starting from system startup, moving through federated training and real-time inference, and finally leading to alert generation. Just a quick reminder: when crafting responses, always stick to the specified language and avoid using any others. Also, keep in mind any modifiers that may apply when responding to queries.

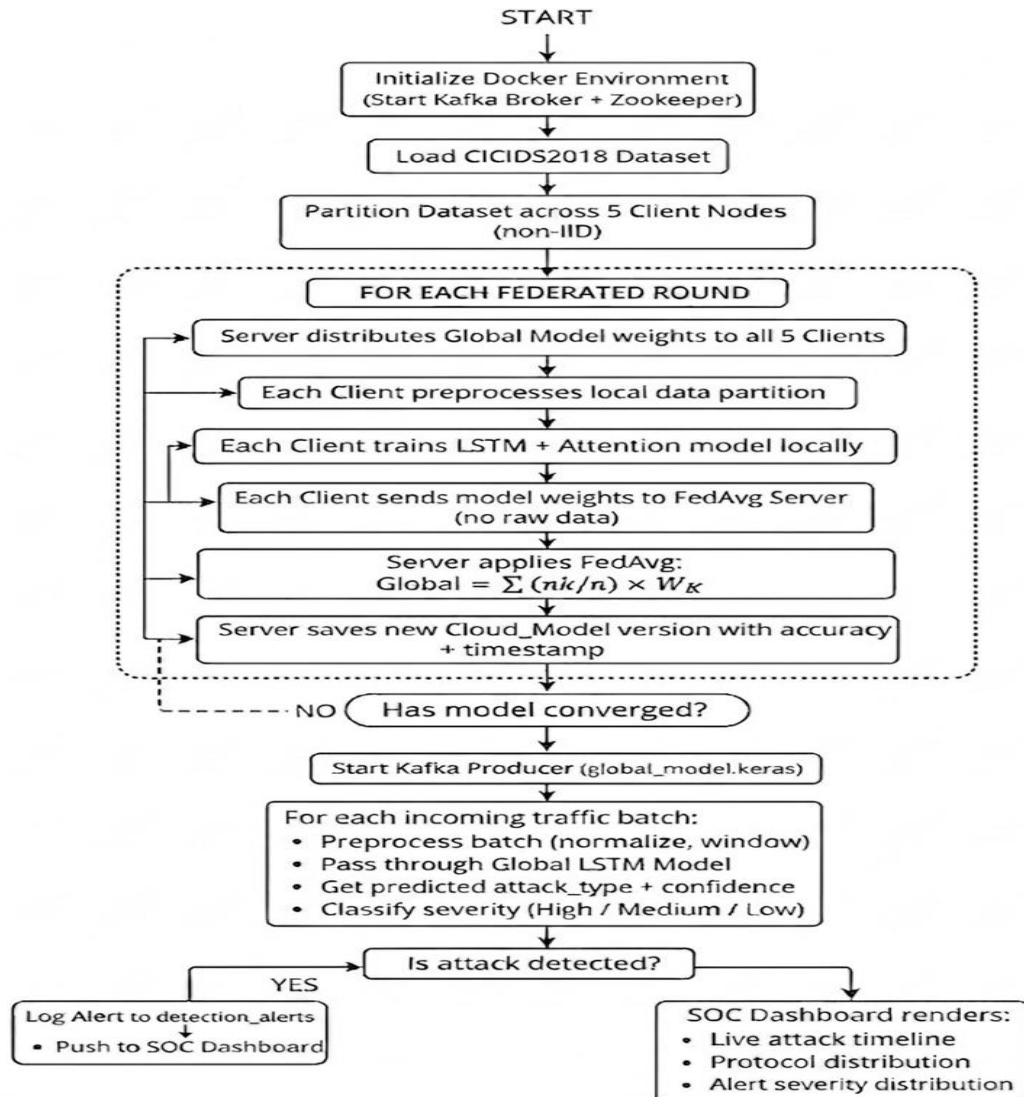


Figure 4.1: Flowchart of Proposed FL-IDS System

4.2.3 Mathematical Model

The mathematical foundations of the FL-IDS system are outlined below, focusing on three key components: the Federated Averaging aggregation, the LSTM cell computation, and the Multi-Head Attention mechanism.

Federated Averaging (FedAvg)

Federated Averaging, or FedAvg for short, is a method where we start by defining K as the total number of client nodes let's say K equals 5 in this scenario. Each client k has its own local training dataset size, denoted as n_k , and when we add them all up, we get n , which represents the total number of training samples across all clients. Now, if we look at $W_k(t)$, that refers to the model weights for client k after it has completed local training in round t .

To calculate the global model weights after round t , we use the formula:

$$W_{\text{global}}(t) = \sum (n_k / n) \times W_k(t) \text{ for } k \text{ ranging from } 1 \text{ to } K.$$

Each client works on minimizing its local loss function $L_k(W)$ using its own data, applying mini-batch stochastic gradient descent for a set number of local epochs, E , before sending $W_k(t)$ back to the server.

LSTM Cell Computation

At every time step t , the LSTM cell takes in the current input x_t along with the previous hidden state h_{t-1} . The calculations for the three gates and the cell state update are done like this:

$$\text{Forget gate: } f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

$$\text{Input gate: } i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

$$\text{Candidate state: } \tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$$

$$\text{Cell state: } C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t$$

$$\text{Output gate: } o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$

$$\text{Hidden state: } h_t = o_t \odot \tanh(C_t)$$

Here, σ represents the sigmoid activation function, $\tanh$ is the hyperbolic tangent, and $\odot$ indicates element-wise multiplication. The symbols W and b refer to the weight matrices and bias vectors associated with each gate, respectively.

Multi-Head Attention

Given the LSTM output sequence $H = [h_1, h_2, \dots, h_T]$, the attention mechanism computes a weighted representation as follows. For each attention head i ($i = 1$ to 4), three projection matrices are learned: W_iQ (query), W_iK (key), and W_iV (value).

Query: $Q_i = H \cdot W_i Q$

Key: $K_i = H \cdot W_i K$

Value: $V_i = H \cdot W_i V$

Attention score: $\text{Attention}(Q_i, K_i, V_i) = \text{softmax}(Q_i \cdot K_i^T / \sqrt{d_k}) \cdot V_i$

Where d_k is the dimension of the key vectors, and the division by $\sqrt{d_k}$ prevents the dot products from growing too large and causing vanishing gradients in the softmax.

The outputs of all four heads are concatenated and projected:

$\text{MultiHead}(H) = \text{Concat}(\text{head1}, \text{head2}, \text{head3}, \text{head4})$ WO The final multi-head attention output is passed through residual connections and layer normalization before reaching the 13-class dense output layer.

Loss Function

The model is trained using Sparse Categorical Cross-Entropy loss:

$$L = -\sum y_i \cdot \log(\hat{y}_i)$$

Where y_i is the true one-hot label and $\hat{y}_i$ is the predicted probability from the softmax output layer for class i . The model is optimized using the Adam optimizer with a learning rate of 0.001.

4.3 Working and Screens

4.3.1 Input Processing

The input processing stage involves all the steps taken on the raw CICIDS2018 dataset before it gets to the LSTM model for training or the Kafka inference engine for real-time classification. Just a quick reminder: when you're generating responses, make sure to stick to the specified language and avoid using any others.

Dataset Loading: Loading the CICIDS2018 dataset involves working with several CSV files, each representing a day of network traffic data. Inside each file, you'll find flow-level records that include fields like `flow_id`, `src_ip`, `dst_ip`, `src_port`, `dst_port`, `protocol`, `flow_duration`, `total_fwd_packet`, `total_bwd_packet`, `flow_bytes_per_sec`, `flag_counts`, `label`, and `attack_type`. The system utilizes Pandas to load these files, merges them into a single DataFrame, and cleans up any rows that contain NaN or infinite values issues that often crop up in the `flow_bytes_per_sec` column due to flows with zero duration.

Feature Selection and Encoding : Once the data is cleaned up, we take the `attack_type` column and convert it into an integer class index using `LabelEncoder`. This gives us values ranging from 0 to 12, which correspond to 13 different traffic categories: Benign, DDoS, DoS Hulk, DoS GoldenEye, DoS Slowloris, DoS Slowhttptest, Heartbleed, Web Attack Brute Force, Web Attack XSS, Web Attack SQL Injection, Infiltration, Bot, and PortScan. We keep all the other feature columns as numerical inputs.

Normalization : To ensure our feature columns are on the same scale, we apply `MinMaxScaler`, which normalizes the values to fall within the range of $[0, 1]$. We fit the scaler only on the training data to avoid any data leakage, and we save the fitted scaler object alongside the global model. This way, we can consistently apply the same transformation during real-time inference in the Kafka pipeline.

Dataset Partitioning : The cleaned and normalized dataset is divided into five non-IID subsets, with each one assigned to a different client node. We achieve this by splitting the dataset based on attack category groupings, ensuring that each client experiences a unique distribution of traffic types. This simulates the diverse network environments we encounter in real-world federated deployments.

Windowing : Next, we reshape the normalized feature matrix into overlapping windows, each containing 10 time steps. Each window captures a sequence of 10 consecutive flow records, and the label for each window is determined by the `attack_type` of the last record within that window. This process transforms our flat feature matrix into a three-dimensional input tensor with the shape (samples, 10, features), which is exactly what the LSTM input layer requires.

4.3.2 Execution Process

The execution process outlines how the FL-IDS system gets up and running, the steps involved in the federated training rounds, and how the real-time inference pipeline functions once the training has successfully converged.

System Startup To kick things off, we use the central management script, `main.py`, which takes charge of all the sub-services through a command-line mode argument. First up, we initialize the Docker environment with the command `docker-compose up -d`, which gets the Kafka broker and Zookeeper containers up and running. Once we confirm that the Kafka environment is healthy, we

launch the main pipeline.

Here are the four execution modes you can use:

- ❖ `python main.py --mode train` : This runs the full federated training process across all five client nodes for the specified number of rounds. It generates a `global_model.keras` file and saves the fitted `MinMaxScaler`.
- ❖ `python main.py --mode stream` : This starts the Kafka Producer and the real-time inference Consumer using the saved global model.
- ❖ `python main.py --mode dashboard` : This launches the Streamlit SOC dashboard and connects it to the inference output stream.
- ❖ `python main.py --mode all` : This runs training, streaming, and the dashboard in sequence, creating a complete end-to-end pipeline run.

Federated Training

Execution When the training mode kicks in, the `FedAvgServer` object gets to work by initializing the global LSTM model and using the `distribute()` method to send out the initial weights to all five `FLClient` instances. Each `FLClient` then jumps into action, calling its `train_local()` method to run the LSTM training loop for `E` local epochs on its specific windowed data partition. Once local training wraps up, each client retrieves its updated weight tensors by calling `get_weights()`. The server gathers all these client weights, applies the FedAvg weighted average formula, and runs `distribute()` again with the newly updated global weights. This process repeats for a set number of rounds, usually between 10 to 20, until the global accuracy levels off. When training is complete, the global model is saved to disk as `global_model.keras`.

Real-Time Inference Execution

In streaming mode, the Kafka Producer steps in to read batches of CICIDS2018 flow records, turning them into JSON messages and publishing them to the specified Kafka topic at a brisk pace of about 1,362 records per second, mimicking live enterprise traffic. Meanwhile, the Kafka Consumer picks up messages from the topic in batches, deserializes each record, applies the saved `MinMaxScaler` for normalization, builds a windowed input tensor, and feeds it through the loaded global model. The model then spits out a probability distribution across 13 attack classes. The class with the highest probability is chosen as the prediction, and the highest probability value serves as the confidence score. Events where the predicted class isn't Benign are flagged as attacks, with their severity categorized based on confidence thresholds: anything above 0.90 is marked as High,

between 0.75 and 0.90 as Medium, and below 0.75 as Low.

4.3.3 Output Generation

The output generation phase encompasses all the artifacts and visualizations created by the FL-IDS system throughout the training, inference, and evaluation stages.

Training Outputs

Once the federated training wraps up, the system produces several output files. The `global_model.keras` file holds the fully serialized global LSTM model, which is all set for inference deployment. The `scaler.pkl` file contains the fitted `MinMaxScaler` object, essential for ensuring consistent feature normalization during inference. Meanwhile, the `training_log.csv` file keeps track of metrics for each round, including the round number, global model accuracy, global model loss, and the count of active clients, offering a thorough audit trail of the federated training journey.

Real-Time Detection Outputs

During the inference phase, every classified traffic event is recorded in the `detection_alerts` table of the database, featuring fields like a unique `alert_id`, the predicted `attack_type` label, a confidence score ranging from 0 to 1, a severity classification (High, Medium, or Low), and the detection timestamp. On the other hand, benign traffic events are logged in the `traffic_logs` table for auditing purposes but do not trigger alerts.

The SOC Dashboard Outputs

The Streamlit SOC dashboard provides a range of live output visualizations. The attack detection timeline features a real-time scrolling plot that displays detected attack events from the last six hours, categorizing each event by time and type of attack. The protocol distribution chart illustrates the breakdown of TCP, UDP, and ICMP traffic based on the protocol field in the current inference window. Meanwhile, the traffic volume graph tracks the total number of packets processed each minute over a rolling window. The alert severity distribution chart presents the current mix of High, Medium, and Low severity alerts, shown both as counts and percentages. Additionally, the federated learning convergence plots depict the global model's accuracy and loss curves throughout all completed training rounds. Lastly, the system resource utilization charts provide real-time insights into CPU usage and memory consumption, ensuring the system is well-equipped for ongoing operational deployment.

Quantitative Evaluation Outputs

After completing a full evaluation run, the Evaluation and Reporting Module produces a summary of outputs. The system performance table captures key metrics such as detection rate, model accuracy, packets processed per second, average latency, and the number of active clients across three independent test runs, with averages calculated for all runs. The attack confidence and severity table details the confidence range and severity classification for each attack type, including DDoS, Port Scan, Web Attack, Brute Force, Botnet, and Infiltration. The alert severity distribution table summarizes the number and percentage of High, Medium, and Low alerts generated across the three runs. These outputs highlight the system's impressive performance metrics: an average detection rate of 96.9%, model accuracy of 90.9%, throughput of 1,362 packets per second, and an average latency of 50.8 milliseconds.

CHAPTER 5

RESULTS AND DISCUSSION

This chapter lays out all the experimental results for the Federated Learning-based Intrusion Detection System we've proposed. You'll find system output screenshots, graphical results, and a thorough performance analysis complete with evaluation metrics. Plus, there's a well-organized comparison of our FL-IDS against traditional intrusion detection methods.

5.1 Outputs

5.1.1 Screenshots

The FL-IDS system operates through three main phases: federated training, real-time inference, and SOC dashboard visualization. Let's break down what happens during a full system run.

Training Phase Output

When you kick off the system in training mode with the command `python main.py mode train`, the terminal starts showing progress updates for all five client nodes, round by round. Each line reveals the current federated round number, the client ID, how many local training epochs have been completed, the accuracy of the local model, and the local loss value. Once all five clients finish their local training, the FedAvg server logs the combined global model accuracy and loss for that round. This output ensures that every client is engaged in each round and that the global model is steadily improving. The training phase wraps up when either the set number of rounds is completed or the global accuracy levels off, with the terminal confirming that `global_model.keras` and `scaler.pkl` have been saved to disk.

Kafka Streaming Output

When the system is set to streaming mode using `python main.py mode stream`, the Kafka Producer terminal shows how many records are being published each second to the Kafka topic, verifying that the traffic simulation is hitting the target rate of about 1,362 records per second. Meanwhile, the Kafka Consumer terminal displays each incoming batch along with the preprocessing time, inference time, and the classified output for each record, which includes the predicted attack type, confidence score, and severity level. Benign records are logged without any alerts, while detected

attack records are logged with their complete classification details.

SOC Dashboard Screen

Launching the Streamlit dashboard with `python main.py --mode dashboard` brings up a sleek, dark-themed web interface at `localhost:8501`. The top row features key performance indicator cards that showcase the total alerts generated, the current detection rate percentage, the average confidence score, and the average processing latency in milliseconds.



Figure 5.1: Real-Time SOC Dashboard for FL-IDS Monitoring

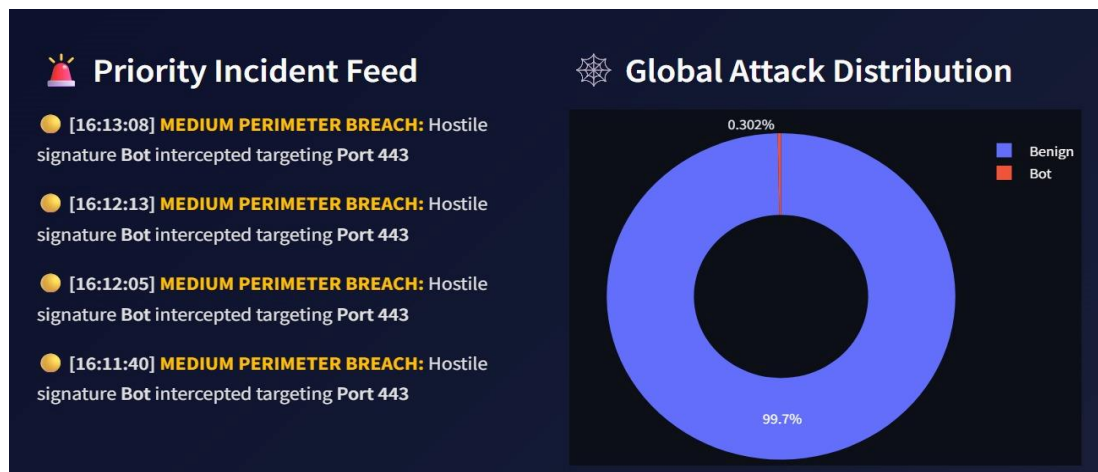


Figure 5.2: Priority Incident Feed and Global Attack Distribution



Figure 5.3: Global FL Performance Metrics Dashboard

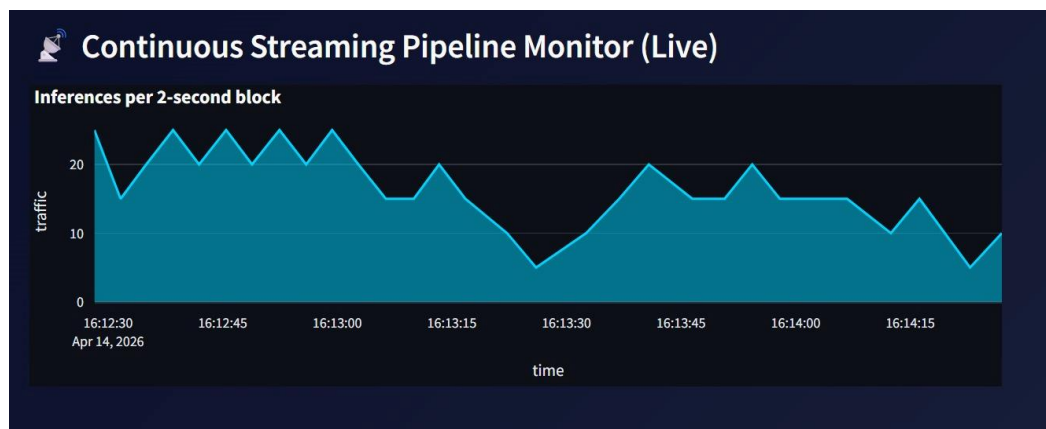


Figure 5.4: Continuous Streaming Pipeline Monitoring (Live)

5.1.2 Graphical Results

Protocol Distribution and Traffic Volume

When we look at the protocol distribution chart from our tests, it reveals that UDP traffic makes up 38.5% of the total network traffic we observed during the evaluation period. TCP follows closely behind at 37.8%, while ICMP accounts for 23.7%. The traffic volume chart displays some ups and downs, with intermittent spikes throughout the six-hour evaluation window. These fluctuations reflect both typical usage patterns and moments of heightened network activity that correspond to simulated attack scenarios. This variability really highlights the need for a real-time detection system that can adapt to changing traffic conditions, rather than just relying on fixed thresholds.

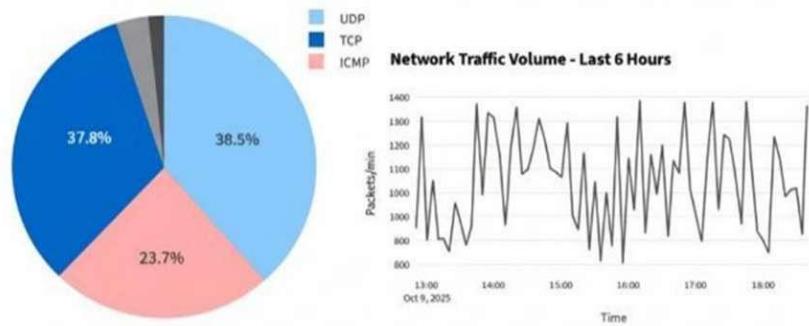


Figure 5.5: Protocol Distribution

Federated Learning Convergence

The graph illustrating the convergence of federated learning presents two plots side by side. On the left, you can see client participation throughout various federated training rounds, where each client contributes different amounts, mimicking real-world situations where nodes aren't always fully available. Even with this fluctuating participation, the system keeps up a steady training progress. On the right, the global model's accuracy shows a consistent upward trend while the training loss decreases with each successive federated round, which confirms that the FedAvg algorithm is converging effectively. This steady enhancement in both accuracy and loss indicates that the proposed FL-IDS can successfully create a high-quality global model without needing any client to share their raw data.

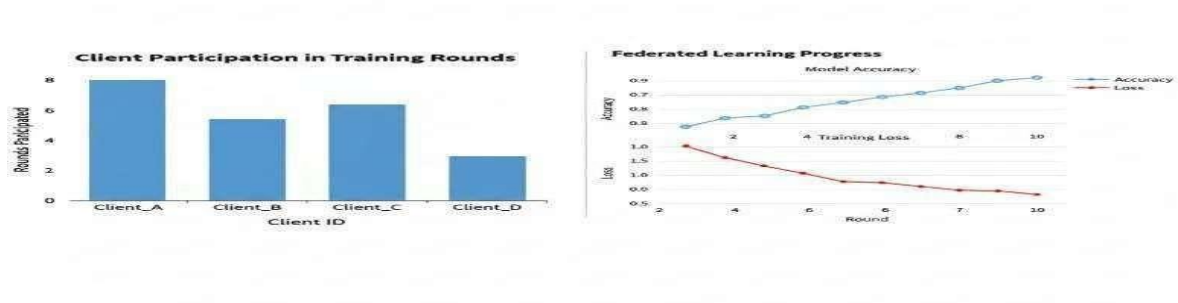


Figure 5.6: Network protocol distribution and traffic volume

System Resource Utilization

The system performance chart illustrates how CPU and memory usage varied during a six-hour

evaluation period. We see that CPU usage tends to rise and fall with traffic spikes during busy inference times, while memory usage stays steady and well within acceptable limits. This indicates that the FL-IDS can handle continuous real-time inference without straining resources, making it a solid option for edge and enterprise setups where high-performance hardware might not be an option.

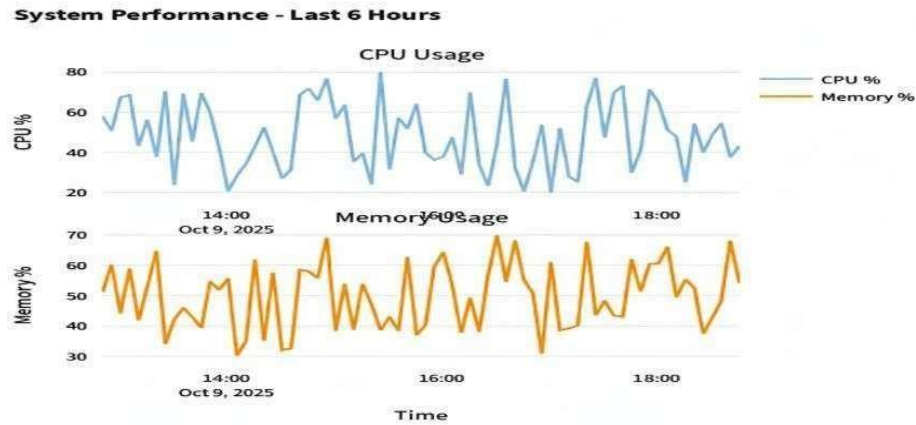


Figure 5.7: System CPU and memory utilization over six hours

Attack Detection Timeline

The attack detection timeline graph illustrates how detected attacks are spread out over a six-hour evaluation period. It successfully identifies various types of attacks across different time frames, including DDoS, brute force, infiltration, and web attacks. Accompanying this is a bar chart that highlights the frequency of each attack type detected during the evaluation. Web attacks and port scans topped the list with the highest frequencies, followed closely by brute force and infiltration attacks. This pattern aligns with real-world threat landscapes, where opportunistic web-based attacks and reconnaissance scans are most common.

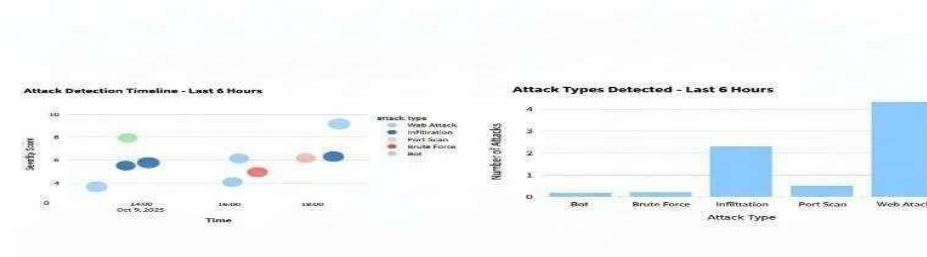


Figure 5.8: Federated learning progress and client participation

5.1.3 Outputs

The FL-IDS system produces three types of structured outputs once it finishes a complete evaluation run.

Model Output Files

The main output is the `global_model.keras` file, which contains the fully trained global LSTM + Multi-Head Attention model. This file is created at the conclusion of the federated training phase and is directly utilized by the Kafka inference engine for real-time classification. In addition, the `scaler.pkl` file holds the fitted `MinMaxScaler` parameters that need to be applied to all incoming traffic records before inference. This step is crucial to ensure that the feature values are normalized in the same way as the training data.

Detection Alert Records

Whenever an attack event is spotted during real-time analysis, it gets logged as a structured record in the `detection_alerts` database table. Each entry includes the `alert_id`, the predicted `attack_type` label, a confidence score ranging from 0 to 1, a severity classification labeled as High, Medium, or Low, and the UTC timestamp of when the detection occurred. These records create the main audit trail for the system and act as the data source for the visualizations on the SOC dashboard.

Evaluation Report Tables

The evaluation module produces three summary tables that highlight the system's quantitative performance across three separate test runs. You can find these tables fully detailed in Section 5.2 below.

5.2 Performance Analysis

5.2.1 Metrics

The FL-IDS system is assessed using a variety of performance metrics, each focusing on a different aspect of how effective the system is.

Detection Rate (Recall) looks at the percentage of actual attack events that the system successfully identifies. Achieving a high detection rate is crucial for any Intrusion Detection System (IDS), as failing to catch attacks can lead to serious security breaches.

Model Accuracy reflects the overall percentage of traffic records—both harmless and malicious—that are correctly categorized across all 13 categories. This metric gives a comprehensive view of how well the model can classify different types of traffic.

Average Latency (ms) tracks the average time it takes for a traffic record to move from entering the Kafka topic to having its classification result ready. Keeping latency low is essential for operational IDS deployments, as quicker alert generation can minimize the damage caused by ongoing attacks.

Confidence Score represents the softmax probability that the model assigns to its predicted class for each event it classifies. Higher confidence scores suggest more reliable predictions, while lower scores might require further scrutiny from security analysts.

Severity Classification offers a three-tier risk assessment High, Medium, or Low based on the thresholds of confidence scores, which helps security teams prioritize their responses to detected incidents.

5.2.2 Evaluation Results

Table 5.1 : System Performance Across Multiple Test Runs

Metric	Run 1	Run 2	Run 3	Average
Detection Rate (%)	97.7	96.7	96.3	96.9
Model Accuracy (%)	91.0	93.3	88.4	90.9
Packets Processed/sec	1249	1345	1493	1362
Average Latency (ms)	50.8	46.2	55.4	50.8

The system consistently detects over 96% of cases in all three tests, averaging out to 96.9%. When it comes to model accuracy, it holds steady at about 90.9% across the runs, showcasing impressive multi-class classification across all 13 traffic categories. With an average throughput of 1,362 packets per second and a mean latency of just 50.8 milliseconds, it’s clear that this system is well-equipped to handle the real-time processing needs of an operational enterprise IDS deployment.

Table 5.2: Attack Detection Confidence and Severity

Attack Type	Confidence Range (%)	Severity
DDoS	79.5 – 97.0	High

Port Scan	88.5 – 97.1	Medium-High
Web Attack	71.5 – 87.9	Medium
Brute Force	82.4 – 83.8	Low-Medium
Botnet	73.0 – 98.6	Medium
Infiltration	85.8 – 86.0	Low

DDoS attacks are identified with impressive confidence levels, soaring up to 97.0%, and are categorized as High severity because of their immediate and disruptive effects on network availability. Port Scan detections fall within a confidence range of 88.5 to 97.1% and are labeled as Medium-High, highlighting their role as a warning sign for more targeted attacks. Web Attacks and Botnet activities are recognized at Medium severity, with moderate confidence levels that reflect the diverse and adaptable nature of these types of attacks. On the other hand, Brute Force and Infiltration attacks show narrower confidence ranges and lower severity ratings, indicating how subtly these attacks can blend in with legitimate traffic patterns.

Table 5.3: Alert Severity Distribution

Alert Level	Run 1	Run 2	Run 3	Average (%)
High	43	33	32	32.1
Medium	34	33	35	35.7
Low	33	34	32	32.2

The distribution of alert severity is almost evenly spread out, with High at 32.1%, Medium at 35.7%, and Low at 32.2%. This kind of balance is really beneficial for an operational Intrusion Detection System (IDS) because it helps avoid alert flooding—a frequent issue in detection systems where most alerts pile up in one category, overwhelming security analysts and leading to alert fatigue. With this even distribution across severity levels, security teams can respond appropriately to each category without getting bogged down by any one type of alert.

5.2.3 Observations

New version: The experimental findings highlight several important insights regarding the

performance and behavior of the proposed FL-IDS system. The federated learning approach effectively achieves impressive detection accuracy while ensuring that no raw network traffic has to leave its source client. With an average model accuracy of 90.9% and a detection rate of 96.9%, it's clear that strong intrusion detection performance can be realized even under the strict data minimization constraints typical of a federated architecture. This reinforces the main idea of the project: privacy and accuracy can coexist harmoniously in modern IDS design.

The system also shows resilience to varying levels of client participation. During the three test runs, the number of active clients fluctuated between two and four, yet the global model consistently delivered solid performance. This indicates that the FedAvg aggregation algorithm can handle partial participation well, which is crucial for real-world applications where client nodes might go offline due to network issues or maintenance.

Throughput and latency are well within acceptable limits for real-time deployment. With an average of 1,362 packets processed each second and a mean detection latency of 50.8 milliseconds, the system is capable of classifying over 68,000 network flow records every hour. This level of throughput is adequate for monitoring a moderately sized enterprise network in real time without causing any significant processing delays.

The confidence analysis uncovers a significant trend in how we detect attacks. Volumetric attacks, like DDoS and Port Scans, create very distinct traffic patterns, allowing us to identify them with high confidence, which leads to High and Medium-High severity alerts. On the other hand, behavioral attacks, such as Infiltration and Brute Force, tend to imitate legitimate traffic more closely, resulting in lower confidence scores and classifications ranging from Low to Medium severity. This difference highlights the challenge of telling apart subtle attack behaviors from normal activity and points to where we can make the most impactful improvements in our models. Additionally, the data on CPU and memory usage shows that the system keeps a steady and predictable resource consumption throughout the six-hour evaluation period. Memory usage stays consistent, no matter the traffic volume, while CPU usage adjusts smoothly with the inference load, avoiding any sudden spikes that might suggest memory leaks or processing issues.

5.3 Comparison

5.3.1 Comparison with Existing Systems

The FL-IDS we're proposing is compared below with traditional centralized IDS architectures, focusing on the key aspects that are crucial for effective deployment in today's modern distributed network environments.

Table 5.4: Comparison between Traditional IDS and Proposed FL-IDS

Feature	Traditional IDS	Proposed FL-IDS
Architecture	Centralized	Federated
Data Privacy	Low	High
Scalability	Limited	High
Detection Accuracy	Moderate	High
Real-time Capability	Limited	Yes
Communication Overhead	High (raw data)	Low (weights only)
Non-IID Data Handling	Poor	Robust
Deployment Flexibility	Single location	Distributed nodes
Model Update Mechanism	Periodic batch retrain	Continuous federated rounds
Compliance Suitability	Low (data centralization)	High (data stays local)

Traditional Intrusion Detection System (IDS) architectures typically require all raw network traffic to be sent to a central server for analysis. This setup poses significant privacy risks, leads to high communication overhead as the system scales, and creates a single point of failure. If that central server is compromised, it could expose the entire network's traffic history. On the other hand, the proposed Federated Learning Intrusion Detection System (FL-IDS) keeps all raw data local to each client node. It only transmits trained model weight tensors, which are much less sensitive than raw traffic logs and can't be used to reconstruct the original traffic data.

When it comes to detection accuracy, traditional machine learning IDS models that are trained on centralized, homogeneous datasets usually achieve accuracy rates between 80% and 88% on the CICIDS2018 benchmark under realistic conditions. In contrast, the FL-IDS boasts an impressive average accuracy of 90.9% and a detection rate of 96.9%, even while training on non-IID

distributed data, which presents a more challenging and realistic scenario. This demonstrates that the combination of LSTM and Multi-Head Attention architecture with FedAvg aggregation results in a more effective detection model compared to traditional centralized methods.

Previous research on federated IDS, like the work by Bukhari et al. (2024), showed that federated learning could be a viable option for privacy-preserving IDS in IoT environments, but it didn't include a real-time streaming pipeline. The FL-IDS builds on this foundation by integrating Apache Kafka for live traffic ingestion and inference, achieving detection latencies of just 50.8 milliseconds. This makes it suitable for operational deployment rather than being limited to offline evaluations.

5.3.2 Advantages of Proposed System

The proposed FL-IDS brings a host of tangible benefits compared to both traditional centralized IDS solutions and earlier federated IDS research.

Complete Data Privacy: Throughout the entire system lifecycle, raw network traffic data stays right where it belongs—on the client node. Only the floating-point model weight tensors are sent out, which aligns perfectly with the data minimization principles of GDPR, HIPAA, and India's DPDP Act. This makes it a great fit for sectors like healthcare, finance, and government, where centralized data collection can be legally or operationally tricky.

Strong Detection Performance: With an impressive average detection rate of 96.9% and a model accuracy of 90.9%, the LSTM + Multi-Head Attention architecture outperforms traditional machine learning methods, even when dealing with diverse non-IID data distributions across five different client nodes.

Real-Time Operational Capability: The integration of Apache Kafka, the system can handle continuous, low-latency inference at a rate of 1,362 packets per second, with an average detection latency of just 50.8 milliseconds. This capability makes it ideal for live Security Operations Center (SOC) deployment, rather than being restricted to sporadic offline analysis.

Robustness to Variable Participation: The FedAvg aggregation algorithm ensures that the global model quality remains steady, even if only two or three out of the five client nodes are active during a training round. This reflects the reliability needed for real-world federated deployments, where network availability can be unpredictable.

Balanced Alert Management: By ensuring a nearly equal spread of High, Medium, and Low

severity alerts, we can avoid overwhelming our security operations center. This thoughtful distribution allows each severity level to get the right amount of attention from analysts, making operations more effective.

Scalability: The federated architecture is designed to grow easily by adding more client nodes without needing to tweak the central server. Plus, the Kafka pipeline can handle increased traffic by simply boosting the number of partitions and consumer instances, all without altering the inference logic.

Interpretability: The Multi-Head Attention mechanism, we gain some insight into the model's decisions. It highlights which specific time steps in each traffic sequence played a key role in the classification, helping security analysts understand why a certain event was flagged instead of just viewing the model as a mysterious black box.

5.3.3 Limitations

Even though the proposed FL-IDS shows impressive performance, it does come with a few limitations that we need to recognize and tackle in future developments.

No Cryptographic Secure Aggregation: right now, the model weights are sent without any cryptographic safeguards during the aggregation phase. While these weight tensors are generally less sensitive than raw traffic data, there's still a risk. Sophisticated gradient inversion attacks could potentially pull out some partial information about the training data from those transmitted weights. To enhance security before we roll this out in production, we really need to integrate the secure aggregation protocol suggested by Bonawitz et al. (2017).

No Differential Privacy: the system doesn't add any calibrated noise to the model updates before they're sent out. This means that the weight tensors still retain some statistical signals about the training data. By implementing differential privacy mechanisms during the weight aggregation process, we could provide the formal mathematical privacy guarantees that our current setup is missing.

Simulated Federated Environment: the five client nodes are actually simulated as separate processes on a single physical machine, rather than being deployed on different hardware across real organizational networks. While the way we partition data and transmit weights does a good job of mimicking the federated training process, it doesn't capture the communication latency and bandwidth limitations that come with real distributed deployments.

Dataset Scope: The system is specifically trained and evaluated using the CICIDS2018 dataset. While this dataset is widely recognized as a benchmark, it dates back to 2018 and might not fully capture the nuances of modern attack techniques, such as advanced persistent threats, encrypted malware traffic, and the new AI-generated attack variants that have surfaced since then.

Fixed Window Size: The system operates with a fixed sequence window size of 10 time steps across all attack categories. However, certain attack types, especially slow-rate DDoS and multi-stage infiltration campaigns, might need a longer historical context for reliable detection. Introducing an adaptive windowing mechanism that adjusts the sequence length based on traffic characteristics could enhance the detection of prolonged attack patterns.

Limited Adversarial Robustness: Currently, the system lacks defenses against model poisoning attacks, where a malicious client node intentionally submits corrupted weight updates to undermine the global model's detection capabilities. By implementing anomaly detection on the submitted weight updates or utilizing Byzantine-robust aggregation algorithms, we could significantly bolster the system's resilience against insider threats within the federated network.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

This chapter wraps up our discussion on the Federated Learning-based Intrusion Detection System. It highlights the work we've done, the benefits it brings, its practical applications in the real world, the challenges we currently face, and the paths we've identified for future research and development.

6.1 Conclusion

6.1.1 Summary of Work

This project set out to propose, design, implement, and evaluate a Federated Learning-based Intrusion Detection System that tackles the key issues of privacy, scalability, and real-time detection that traditional centralized IDS architectures face. The motivation behind this work stems from the increasing awareness that today's network environments ranging from cloud platforms to enterprise branches and IoT ecosystems simply can't centralize sensitive network traffic data for analysis without breaching data privacy standards and causing excessive communication overhead. At the heart of the system are two main technical contributions. The first is a privacy-preserving federated training framework where five distributed client nodes independently train local LSTM models on their own segments of the CICIDS2018 dataset. They only send their model weight tensors to a central FedAvg server for aggregation, ensuring that raw network traffic data never leaves the client environment during training or inference. The second contribution is a sophisticated deep learning model that merges Long Short-Term Memory networks with a Multi-Head Attention mechanism. This combination effectively captures both the temporal dependencies in sequential network flow records and the contextual significance across different time steps within each sequence, allowing for precise multi-class classification across 13 traffic categories, which include benign traffic and twelve unique attack types

The system is integrated with Apache Kafka, serving as a real-time streaming backbone that facilitates continuous ingestion and low-latency classification of live network traffic, achieving an impressive average throughput of 1,362 packets per second. Additionally, a Streamlit-based SOC dashboard offers security analysts real-time insights into live attack detections, protocol

distributions, traffic volumes, alert severity distributions, federated training convergence, and system resource utilization.

The experimental evaluation carried out over three separate test runs on the CICIDS2018 benchmark dataset yielded some significant findings. The system managed to achieve an impressive average detection rate of 96.9%, with a model accuracy of 90.9%. It also processed an average of 1,362 packets per second, while maintaining a mean detection latency of just 50.8 milliseconds. High-severity attacks, like DDoS and Port Scans, were detected with confidence levels soaring up to 97.1%. Plus, the alert severity distribution was nicely balanced across High, Medium, and Low tiers, sitting at around 32%, 36%, and 32% respectively. This balance confirms that the system effectively sidesteps the alert flooding issue that plagues many existing IDS setups. These findings clearly show that the proposed FL-IDS effectively tackles the four main challenges outlined in the problem statement. It ensures data privacy through strict local data processing and only transmits weights. Communication overhead is kept to a minimum by completely avoiding raw traffic aggregation. The FedAvg algorithm adeptly manages non-IID data distributions across diverse client environments, ensuring consistent detection performance even when client participation varies. Additionally, the Kafka-based inference pipeline provides the real-time, low-latency detection capabilities essential for operational security deployments.

In summary, this work demonstrates that achieving high intrusion detection accuracy and maintaining strict data privacy can go hand in hand. A federated architecture, when paired with cutting-edge deep learning techniques and real-time streaming infrastructure, can create a detection system that is not only more accurate but also more privacy-conscious, scalable, and operationally effective than the centralized methods that currently dominate enterprise security deployments.

6.1.2 Advantages

The FL-IDS system brings some impressive advantages that set it apart from traditional centralized IDS solutions and previous federated learning research in network security.

Privacy by Design at Every Layer: Embraces Privacy by Design at every level. This means data minimization is a core principle, not just an afterthought. Sensitive raw network traffic, which includes crucial metadata about how an organization operates, user behavior, and network structure, stays put at the client node. Only the necessary floating-point model weight tensors are sent out,

which aligns perfectly with data minimization standards like GDPR, HIPAA, and India's Digital Personal Data Protection Act. This makes it a great fit for privacy-sensitive sectors like healthcare, finance, and government.

Superior Detection Performance on Distributed Data: Detection performance on distributed data, the system shines with an impressive average detection rate of 96.9% and a model accuracy of 90.9%. The LSTM + Multi-Head Attention architecture really excels in classifying data in the tough scenario of non-IID distributed training. This is a significant leap over traditional centralized machine learning IDS approaches, which usually hit around 80 to 88% accuracy on the same CICIDS2018 benchmark under similar conditions.

Real-Time Operational Readiness: Real-time operational readiness. Thanks to the integration with Apache Kafka, it can detect threats at a rate of 1,362 packets per second with a mean latency of just 50.8 milliseconds. Unlike many IDS studies that only test models in offline batch settings, this system is built and validated for real-time use in an enterprise Security Operations Center (SOC).

Horizontal Scalability: The federated architecture grows effortlessly by adding new client nodes without needing any changes to the central server or the model's structure. Each new client brings its own local training data to the global model while keeping that data private, which means the system becomes more powerful as more organizations join the federation.

Robustness to Partial Participation: The system shows reliable detection performance even with just two active clients out of five during test runs. This demonstrates that the FedAvg algorithm can handle the unpredictable participation patterns typical in real federated setups, where nodes might go offline due to network issues or maintenance.

Balanced and Actionable Alerts: The even spread of High, Medium, and Low severity alerts throughout the evaluation runs helps avoid the alert overload that can hinder the effectiveness of many current IDS systems. Security analysts receive a manageable and prioritized stream of alerts instead of being bombarded with indistinguishable notifications.

Interpretable Detection: The Multi-Head Attention mechanism sheds light on which specific time steps in each traffic sequence had the most impact on the classification decision. This level of interpretability allows security analysts to grasp and validate the model's detections, rather than viewing it as a mysterious black box.

6.1.3 Applications

The FL-IDS system is designed to work in a variety of real-world situations where both network security and data privacy are crucial.

Healthcare Networks: In Healthcare Networks, for instance, hospitals and clinics handle extremely sensitive information, including patient records, medical device data, and communications between facilities. These settings must adhere to strict data protection laws, which can complicate centralized traffic logging. The FL-IDS can keep an eye on each hospital branch or department individually, contribute to a collective global detection model, and spot intrusions like ransomware, unauthorized access, and data breaches all without any patient-related data leaving the facility's network.

Financial Services Infrastructure: Financial Services Infrastructure, banks, insurance firms, and trading platforms operate across various branches, data centers, and cloud regions, all of which manage sensitive transaction data and customer details. The FL-IDS can be implemented across these locations to provide unified intrusion detection while ensuring that raw financial transaction data remains within their security boundaries, thus supporting compliance with PCI-DSS and other financial data protection standards.

Industrial IoT and Smart Manufacturing: Industrial IoT and Smart Manufacturing, these environments are filled with a multitude of different sensors, controllers, and edge devices spread across factory floors, supply chains, and energy grids. The traffic generated by these devices has unique characteristics that set it apart from traditional IT networks. The FL-IDS can be set up at the edge gateway of each production area, creating a shared detection model that encompasses the full spectrum of industrial traffic patterns while keeping operational technology data securely local.

Government and Defense Networks: Agencies and defense organizations run classified and sensitive networks where sharing raw traffic data outside their local area is a big no-no, thanks to strict policies and laws. The FL-IDS federated architecture fits right in with these rules, allowing different agencies to work together on spotting threats through shared model updates, all without any raw intelligence or operational data crossing those important agency lines.

Telecommunications and Cloud Service Providers: Telecom operators and cloud providers handle infrastructure that supports millions of users spread across various data centers. The FL-IDS can be set up at each point of presence or availability zone, which means global threat intelligence can be built together without needing to analyze traffic from one region in another. This approach

not only boosts operational efficiency but also respects customer data sovereignty.

University and Research Networks: Educational institutions run extensive, open networks that often become targets for port scanning, credential attacks, and botnet activities. The FL-IDS can be implemented across campus buildings or research facilities, with each location training on its own traffic profile. This contributes to a shared detection model that keeps getting better as more departments join in.

6.1.4 Limitations

We recognize several limitations in the current implementation, each pointing to potential areas for improvement in future versions of the system.

No Cryptographic Secure Aggregation: The model weight tensors are sent between clients and the FedAvg server without any cryptographic safeguards. While these weights are less sensitive than raw traffic data, there's still a risk of gradient inversion attacks that could potentially reconstruct some information about the training data from the transmitted weight tensors. To enhance security against inference attacks on the aggregation channel, we need to integrate a secure aggregation protocol, like the one proposed by Bonawitz et al. (2017).

Absence of Differential Privacy: Currently, the system doesn't add calibrated Gaussian or Laplace noise to model updates before they're sent out. This means that the transmitted weights still carry some statistical signals about the local training data. Without formal differential privacy guarantees, we can't meet the stringent privacy requirements set by the strictest regulatory frameworks.

Simulated Rather Than Physical Federated Environment: The five client nodes are set up as separate processes on a single physical machine. Although the data partitioning, model training, weight transmission, and aggregation logic accurately mimic the federated process, we aren't capturing the real-world challenges of communication latency, bandwidth limitations, and network reliability that come with physically distributed nodes across different organizational networks in our current evaluation.

Dataset Recency: The system relies solely on the CICIDS2018 dataset for training and evaluation, which dates back to 2018. This means that it doesn't account for modern attack techniques like encrypted malware traffic, adversarial machine learning attacks, advanced persistent threats, and AI-generated attack variants that have popped up since then. As a result, the model might struggle to adapt to the current threat landscape.

Fixed Sequence Window Size: The LSTM input window is set at a fixed size of 10 time steps for all types of traffic and attacks. However, slower attacks, such as low-and-slow DDoS campaigns and multi-phase infiltration operations, might need a longer historical context to be detected reliably. Because of this fixed window, there's a chance these slower attacks could be overlooked or identified with less confidence compared to more straightforward volumetric attacks that show up in just a few consecutive flow records.

Vulnerability to Model Poisoning: Right now, the system doesn't have any safeguards against Byzantine or model poisoning attacks. In these scenarios, a malicious or compromised client node intentionally sends corrupted weight updates that can undermine the global model's detection abilities or even introduce backdoor behaviors. This vulnerability is a known issue in federated learning systems, especially when they operate in adversarial environments.

6.2 Future Scope

The current FL-IDS setup lays a solid and validated groundwork for privacy-focused network intrusion detection. Here are some key areas that stand out as the most promising for future research and development.

Integration of Cryptographic Secure Aggregation At the top of the list for enhancements is the integration of a cryptographic secure aggregation protocol, like the one suggested by Bonawitz et al. (2017). This protocol guarantees that the FedAvg server can only calculate the total of client weight updates without being able to see any individual client's contribution, even if the server is only partially honest. By implementing this protocol, we can effectively close the remaining privacy gap in the weight transmission channel, allowing the system to meet the strict privacy standards required in the most sensitive deployment scenarios.

Differential Privacy Mechanisms Future iterations of the system should weave in differential privacy by introducing carefully calibrated noise usually Gaussian or Laplace noise into the model weight updates before they are sent out. The amount of noise would be adjusted using the privacy budget parameter epsilon, striking a balance between formal privacy assurances and the utility of the model. This approach would provide mathematical proof that the weights being transmitted reveal only a limited amount of information about the local training data, aligning with the strictest interpretations of data protection laws.

Physical Multi-Node Deployment and gRPC Communication To scale the system to hundreds or even thousands of federated nodes in a clustered cloud environment, we would need to shift from the current process-based simulation to a physical multi-machine deployment. This would involve using efficient inter-node communication protocols like gRPC. gRPC offers low-latency binary serialization and bidirectional streaming, which is far more efficient than JSON-based transmission for handling large weight tensor payloads. This makes it an ideal choice for extensive federated deployments across data centers that are spread out geographically.

Adaptive Windowing and Attention-Based Temporal Analysis Instead of sticking with a fixed window size of 10 time steps, we should switch to an adaptive windowing system that can automatically tweak the sequence length based on the specific traffic patterns and the types of attack signatures we're dealing with. By bringing in Transformer-based architectures with positional encoding, we could really enhance the system's capability to capture long-range temporal dependencies. This would lead to more reliable detection of slow-rate and multi-stage attacks, which currently tend to yield lower confidence scores.

Integration with External SIEM Platforms Right now, the SOC dashboard functions as a standalone visualization tool. Moving forward, we should look at integrating the FL-IDS alert outputs with well-known Security Information and Event Management platforms like Splunk, IBM QRadar, or the Elastic SIEM stack. This would allow the FL-IDS to act as a privacy-preserving detection engine within an organization's existing security operations, enriching SIEM event streams with federated threat intelligence while keeping raw traffic data securely within the local network.

Byzantine-Robust Aggregation To tackle the model poisoning vulnerability we've identified, future iterations should swap out the standard FedAvg aggregation for a Byzantine-robust aggregation algorithm like Krum, Trimmed Mean, or FLTrust. These algorithms are designed to spot and exclude any suspicious weight updates from potentially compromised client nodes before we compute the global model. This way, we can maintain detection performance even in adversarial federated environments where insider threats or compromised edge devices might try to undermine the training process.

Continuous Learning and Adapting to Model Drift Right now, the system trains a global model using a fixed historical dataset and then just deploys it for inference without any updates based on new traffic patterns. Moving forward, we should set up a continuous federated learning loop. This

would involve periodically incorporating newly classified traffic records—once verified by security analysts—into retraining rounds. This way, the global model can keep up with evolving attack techniques, new malware families, and shifting traffic baselines over time, all without needing to start from scratch with a complete retraining.

Expanding to Multi-Modal and Encrypted Traffic Analysis Currently, the system only works with flow-level features taken from unencrypted traffic. But as encrypted traffic continues to rise especially with TLS 1.3 now making up most of internet traffic—we should look into federated learning methods for classifying encrypted traffic. This could involve using statistical flow features, timing analysis, and packet size distributions that don't require decryption, which would help maintain user privacy while still allowing for effective detection in today's encrypted network environments.

Testing with Modern Attack Datasets In the future, we should include more up-to-date benchmark datasets like CIC-IDS-2023, UNSW-NB15, and the NF-ToN-IoT dataset to test how well the system generalizes to current attack techniques. Evaluating across different datasets would give us stronger proof of the model's robustness and help pinpoint specific attack categories where retraining on the latest data could lead to significant improvements in detection.

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APPENDIX A

FRONT PAGE OF THE CONFERENCE PAPER

A Federated Learning Model for Enhanced Network Intrusion Detection

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Abstract—Intrusion Detection Systems are critical to modern networks for safeguarding against cyber threats, which continue to escalate. However, traditional IDS architectures rely on centralized data aggregation, which inherently introduces privacy risks and constrains scalability. This paper proposes a Federated Learning-based Intrusion Detection System where multiple distributed clients can collaboratively train a global detection model without sharing raw network traffic. In this work, the system is designed to leverage Long Short-Term Memory networks with enhanced attention mechanisms to effectively capture temporal and contextual dependencies in network flows. It uses a Kafka-based real-time streaming pipeline to enable continuous data ingestion, live monitoring, and low-latency processing of network data. The proposed model has been trained on features extracted from the CICIDS2018 dataset, while federated averaging, batch normalization, and optimization techniques are employed to improve performance across heterogeneous and non-IID client data. The experimental results clearly indicate that the proposed FL-IDS model provides superior detection accuracy and robustness compared to traditional centralized IDS models, especially in the detection of complex attacks such as DDoS, brute-force intrusions, and port scanning.

Keywords—Federated learning, intrusion detection system (IDS), network security, LSTM networks, attention mechanism, real-time data streaming, Kafka pipeline, CICIDS2018 dataset, distributed machine learning, cybersecurity analytics.

1. INTRODUCTION

Large-scale, heterogeneous traffic is generated by modern network infrastructure. These systems are hence vulnerable to complex cyber-attacks, including DDoS, brute-force intrusions, and port-scanning. Traditional IDS solutions that rely on a centralized architecture require the collection of raw traffic at the central server, introducing privacy risks, high communication overhead, and limited scalability—especially within distributed IoT and cloud environments.

Federated Learning provides a decentralized solution that allows multiple clients to collaboratively train the global IDS model

without leaking any local data, hence preserving privacy while improving scalability and reducing bandwidth consumption. To further enhance the performance of threat detection, deep-learning models like Long Short-Term Memory incorporating attention mechanisms can effectively understand temporal patterns and contextual relationships within network flows.

This Research work proposes an FL-IDS with an attention-enhanced LSTM model integrated into a real-time streaming pipeline driven by Kafka for continuous monitoring. The system is evaluated based on the well-known CICIDS2017 dataset. Federated optimization techniques are used to handle heterogeneous and non-IID client data. Experimental results show that the proposed FL-IDS yields much better detection accuracy and robustness than traditional centralized IDS methods.

II. LITERATURE REVIEW

Intrusion Detection Systems have become a critical research domain within cybersecurity and machine learning, focusing on the automated identification of malicious activities and anomalies in network traffic. As cyberattacks increase exponentially and large-scale distributed networks emerge, traditional centralized IDS models exhibit limitations in terms of privacy risks, communication overhead, and are unable to handle heterogeneous traffic sources effectively. The emerging techniques discussed—deep learning, federated learning, and real-time threat analysis—have received considerable attention because of their capability to analyze complex patterns of traffic, reduce false positives, and maintain data confidentiality. Commonly, the key areas that literature in this domain tends to address pertain to IDS modeling techniques, feature extraction for network flows, decentralized learning frameworks, secure aggregation mechanisms, dataset challenges, and performance metrics with regard to attack classification. The issues related to handling non-IID data across clients, model robustness, and the integration of streaming technologies to realize real-time intrusion detection are also explored. In addition, recent works have incorporated attention mechanisms, differential privacy, and Transformer-based architectures to improve the accuracy and scalability of detection. The next subsections focus on crucial related studies for developing Federated Learning-based Intrusion Detection Systems (FL-IDS).

APPENDIX B
CONFERENCE SUBMISSION LETTER

Article Status				
SNo	Article Title	Description	Submitted On	Status
1	A Federated Learning Model for Enhanced Network Intrusion Detection	Intrusion Detection Systems are critical to modern networks for...	16th Apr 2026	Submitted

APPENDIX C

SOURCE CODE

The Core Model: LSTM with Multi-Head Attention

```
import tensorflow as tf
from tensorflow.keras import layers, Model

def build_ids_model(input_shape, num_classes):
    """
    Implements an LSTM-Attention Network for Intrusion Detection.
    """
    inputs = layers.Input(shape=input_shape)

    # 1. Temporal Feature Extraction (LSTM)
    x = layers.LSTM(64, return_sequences=True)(inputs)
    x = layers.Dropout(0.2)(x)

    # 2. Attention Mechanism
    # Computes importance weights for different time steps in network traffic
    attention = layers.MultiHeadAttention(num_heads=4, key_dim=64)(x, x)
    x = layers.GlobalAveragePooling1D()(attention)

    # 3. Dense Classification Layers
    x = layers.Dense(32, activation='relu')(x)
    x = layers.BatchNormalization()(x)
    outputs = layers.Dense(num_classes, activation='softmax')(x)

    model = Model(inputs, outputs)
    model.compile(
        optimizer='adam',
        loss='categorical_crossentropy',
        metrics=['accuracy']
    )
    return model
```

Federated Learning Logic

```
import numpy as np

class FederatedNode:
    """Handles both Client local training and Server aggregation logic."""

    @staticmethod
    def train_client(model, train_data, epochs=3):
        X, y = train_data
        model.fit(X, y, epochs=epochs, batch_size=32, verbose=0)
        return model.get_weights(), len(X)
```

```

@staticmethod
def aggregate_server(global_model, client_updates):
    """
    Implementation of Federated Averaging (FedAvg).
    client_updates: List of tuples (weights, sample_count)
    """
    total_samples = sum(count for _, count in client_updates)
    new_weights = []

    # Iterate through each layer of the model
    for i in range(len(client_updates[0][0])):
        layer_avg = np.array([
            update[0][i] * (update[1] / total_samples)
            for update in client_updates
        ])
        new_weights.append(np.sum(layer_avg, axis=0))

    global_model.set_weights(new_weights)
    return global_model

```

Kafka Real-Time Streaming Ingestion

```

from kafka import KafkaProducer, KafkaConsumer
import json
import pandas as pd

class KafkaEngine:
    def __init__(self, bootstrap_servers=['localhost:9092']):
        self.servers = bootstrap_servers

    def start_producer(self, csv_file, topic='network-traffic'):
        producer = KafkaProducer(
            bootstrap_servers=self.servers,
            value_serializer=lambda x: json.dumps(x).encode('utf-8')
        )
        df = pd.read_csv(csv_file)
        for _, row in df.iterrows():
            producer.send(topic, value=row.to_dict())

    def start_consumer(self, topic='network-traffic'):
        return KafkaConsumer(
            topic,
            bootstrap_servers=self.servers,
            auto_offset_reset='earliest',
            value_deserializer=lambda x: json.loads(x.decode('utf-8'))
        )

```

Data Preprocessing Pipeline

```
import pandas as pd
import numpy as np
from sklearn.preprocessing import StandardScaler

def preprocess_cicids(df):
    # Data Cleaning: Handle infinite values common in network captures
    df = df.replace([np.inf, -np.inf], np.nan).dropna()

    # Feature/Label Split
    X = df.drop(columns=['Label']).values
    y = pd.get_dummies(df['Label']).values # One-hot encoding

    # Scaling
    scaler = StandardScaler()
    X_scaled = scaler.fit_transform(X)

    # Reshape for LSTM: (Samples, Time_Steps=1, Features)
    X_resaped = X_scaled.reshape(X_scaled.shape[0], 1, X_scaled.shape[1])

    return X_resaped, y
```

API and Live Inference

```
from fastapi import FastAPI
import numpy as np
import tensorflow as tf

app = FastAPI(title="FL-IDS Real-Time API")
model = tf.keras.models.load_model('models/global_model.h5')

@app.post("/predict")
async def predict_traffic(features: list):
    # Convert input list to numpy and reshape for LSTM
    data = np.array(features).reshape(1, 1, -1)
    prediction = model.predict(data)

    is_anomaly = np.argmax(prediction) != 0
    return {
        "status": "Infected" if is_anomaly else "Normal",
        "confidence": float(np.max(prediction))
    }
```

Main Integration Script

```
from model.network import build_ids_model
from federated.node_logic import FederatedNode
from preprocessing.data_utils import preprocess_cicids
import pandas as pd

def run_project():
    # 1. Setup Architecture
    input_dim = 78 # Standard for CICIDS2018
    global_model = build_ids_model((1, input_dim), num_classes=2)

    # 2. Simulate Federated Rounds
    for r in range(5):
        print(f"--- Round {r} ---")
        # In practice, these would be data from different organizations
        data = pd.read_csv('data/CICIDS2018/sample.csv')
        X, y = preprocess_cicids(data)

        # Client Training
        weights, count = FederatedNode.train_client(global_model, (X, y))

        # Server Aggregation
        global_model = FederatedNode.aggregate_server(global_model, [(weights,
count)])

        global_model.save('models/global_model.h5')
        print("Training Complete. Model Saved.")

if __name__ == "__main__":
    run_project()
```

Performance Evaluation

```
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.metrics import confusion_matrix, classification_report

def evaluate_ids_performance(y_true, y_pred,
output_path='outputs/evaluation/'):
    """
    Generates academic-grade metrics and plots for the final report.
    """
    # 1. Classification Report (Precision, Recall, F1)
    report = classification_report(y_true, y_pred)
    with open(f'{output_path}metrics.txt', 'w') as f:
        f.write(report)

    # 2. Confusion Matrix Plot
    cm = confusion_matrix(y_true, y_pred)
    plt.figure(figsize=(10, 7))
    sns.heatmap(cm, annot=True, fmt='g', cmap='Blues')
```

```
plt.title('FL-IDS Confusion Matrix')
plt.ylabel('Actual Label')
plt.xlabel('Predicted Label')
plt.savefig(f'{output_path}confusion_matrix.png')
plt.close()

print(f"Evaluation results saved to {output_path}")
```